



US012411665B1

(12) **United States Patent**
Mannheimer et al.

(10) **Patent No.: US 12,411,665 B1**
(45) **Date of Patent: Sep. 9, 2025**

(54) **USER INTERFACE FOR MACHINE
LEARNING MODEL GENERATION**

(58) **Field of Classification Search**
CPC . G06F 8/34; G06F 3/0482; G06F 8/60; G06F
2203/04803
See application file for complete search history.

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(56) **References Cited**
U.S. PATENT DOCUMENTS

10,885,452 B1 * 1/2021 Garg G06N 7/01
2014/0040871 A1 * 2/2014 Schwan G06F 11/3664
717/141

(Continued)

OTHER PUBLICATIONS

(SAP, "Automated Analytics User Guides and Scenarios", Aug. 28,
2020) (Year: 2020).*

(Continued)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **17/521,820**

(22) Filed: **Nov. 8, 2021**

Related U.S. Application Data

(60) Provisional application No. 63/242,012, filed on Sep.
8, 2021.

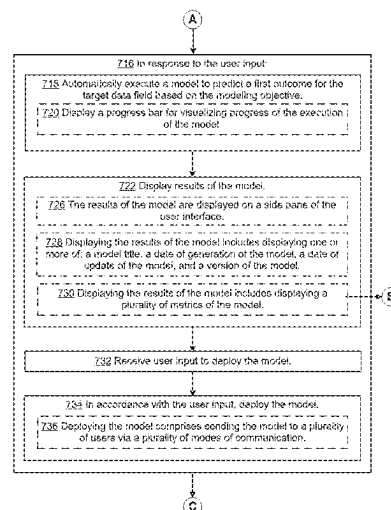
(51) **Int. Cl.**
G06F 8/34 (2018.01)
G06F 3/0482 (2013.01)
G06F 8/60 (2018.01)

(52) **U.S. Cl.**
CPC **G06F 8/34** (2013.01); **G06F 3/0482**
(2013.01); **G06F 8/60** (2013.01); **G06F**
2203/04803 (2013.01)

(57) **ABSTRACT**

A computing device displays, in a user interface, a workflow that includes a plurality of steps. In response to user selection of a first step of the plurality of steps, the computing device displays a list of data sources. The device receives user selection of a first data source of the data sources. The device receives user input specifying a target data field from the first data source and a modeling objective for the target data field. In response to the user input, the device automatically executes a model to predict a first outcome for the target data field based on the modeling objective. The device displays results of the model. The device receives user input to deploy the model. In accordance with the user input, the device deploys the model.

20 Claims, 37 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2016/0232537	A1 *	8/2016	Nonez	G06F 16/24578
2018/0052898	A1 *	2/2018	Allan	G06F 16/2358
2019/0236514	A1 *	8/2019	Schubert	G06F 9/451
2019/0347282	A1 *	11/2019	Cai	G06N 5/022
2020/0134641	A1 *	4/2020	Morgan	G06F 3/04817
2020/0134683	A1 *	4/2020	Boren	G06F 16/904
2020/0174755	A1 *	6/2020	Rose	G06F 8/35
2020/0380432	A1 *	12/2020	Wang	G06F 3/04817
2021/0232920	A1 *	7/2021	Parangi	G06F 16/2457
2021/0319034	A1 *	10/2021	Bern	G06F 16/9538
2021/0406325	A1 *	12/2021	Sinn	G06F 16/9535
2021/0406369	A1 *	12/2021	Healy	G06F 21/562
2022/0050695	A1 *	2/2022	Gajendran	G06N 20/00
2022/0245511	A1 *	8/2022	Perian	G06N 5/04
2022/0309391	A1 *	9/2022	Patel	G06F 16/906
2022/0358286	A1 *	11/2022	Wilson-Thomas ...	G06F 40/284

OTHER PUBLICATIONS

(Clairvoyant, "No-code Machine Learning Model With Azure ML Designer", Aparna Kesarkar, Aug. 27, 2021) (Year: 2021).*

* cited by examiner

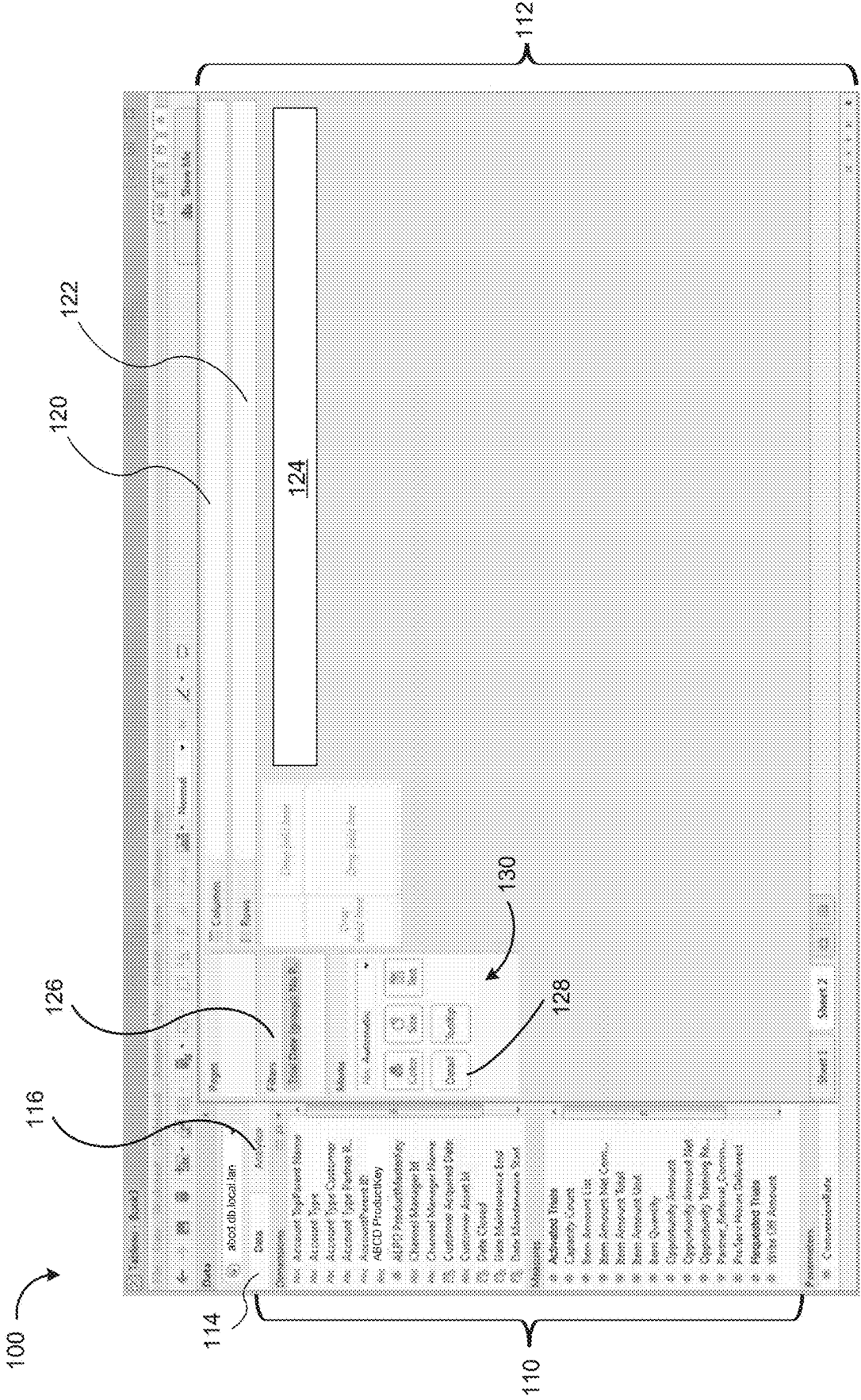


Figure 1

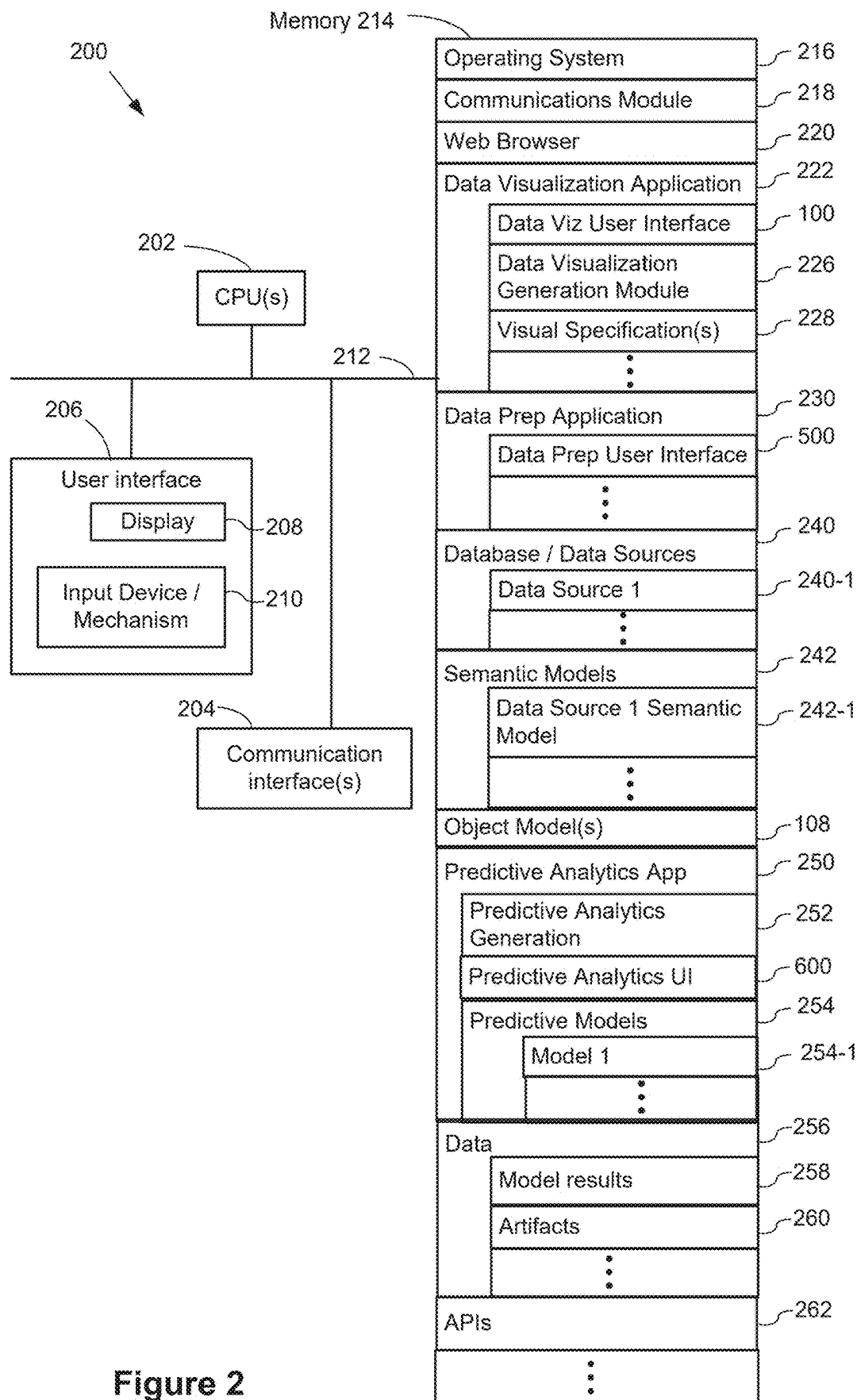


Figure 2

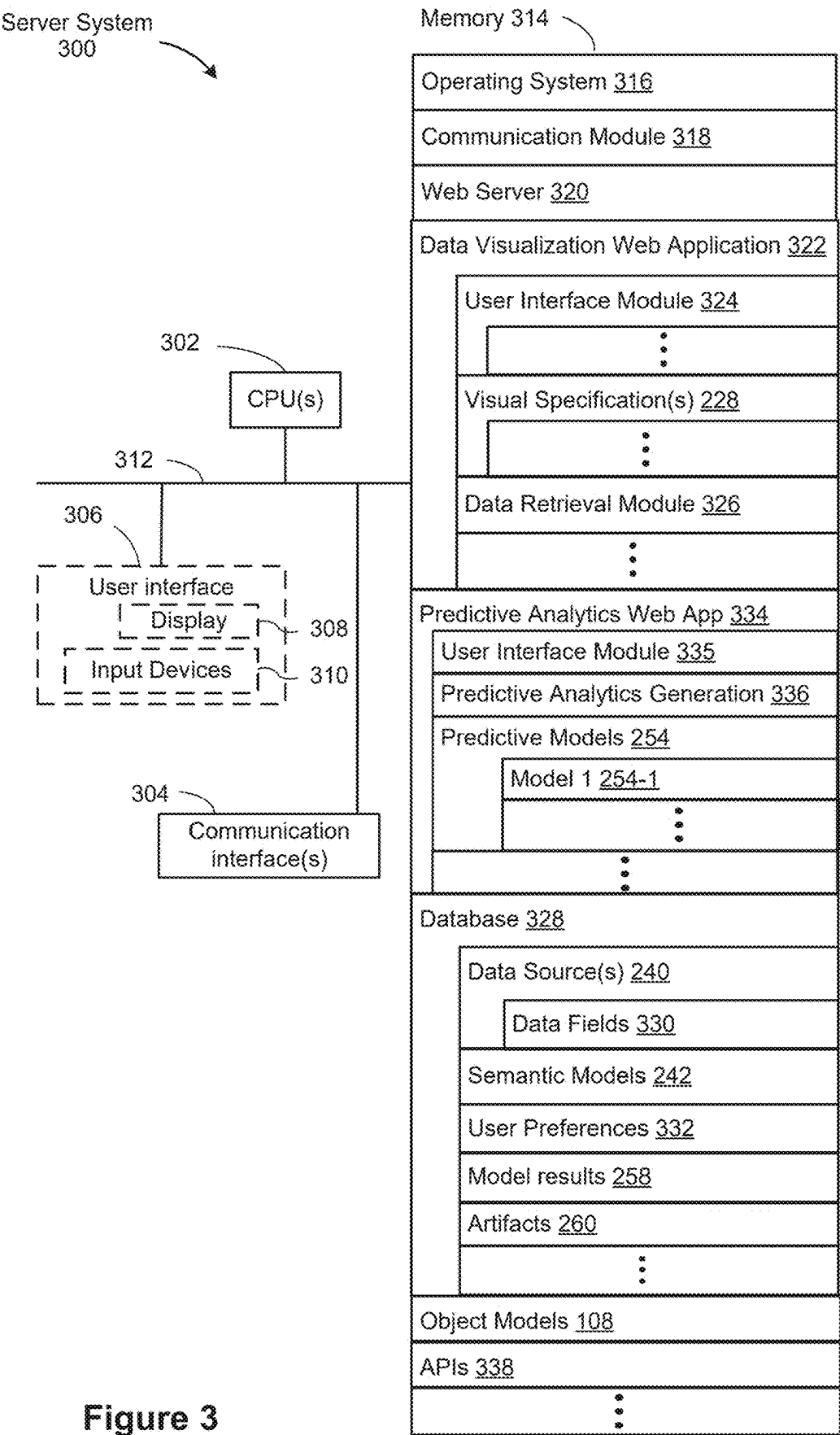
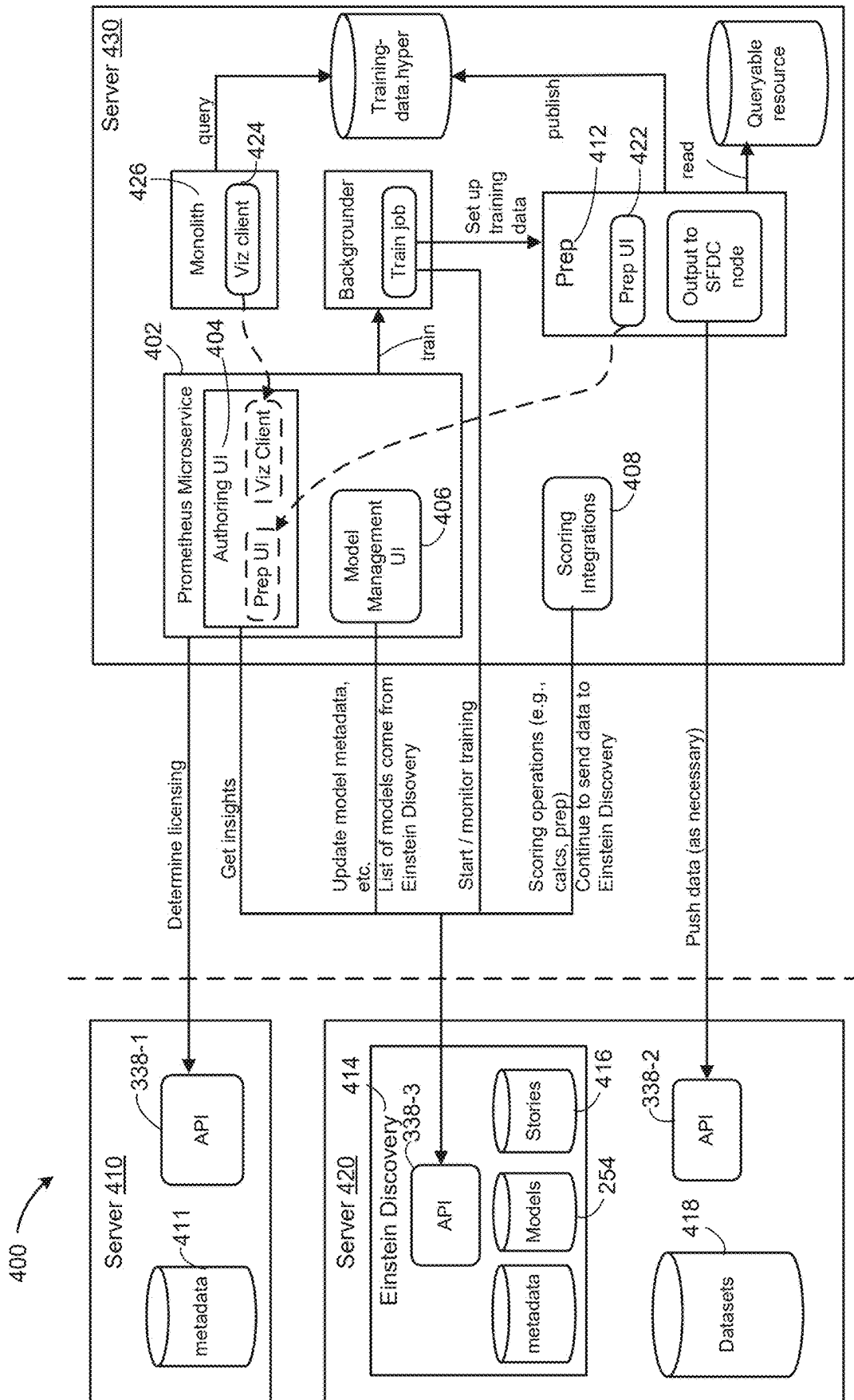


Figure 3



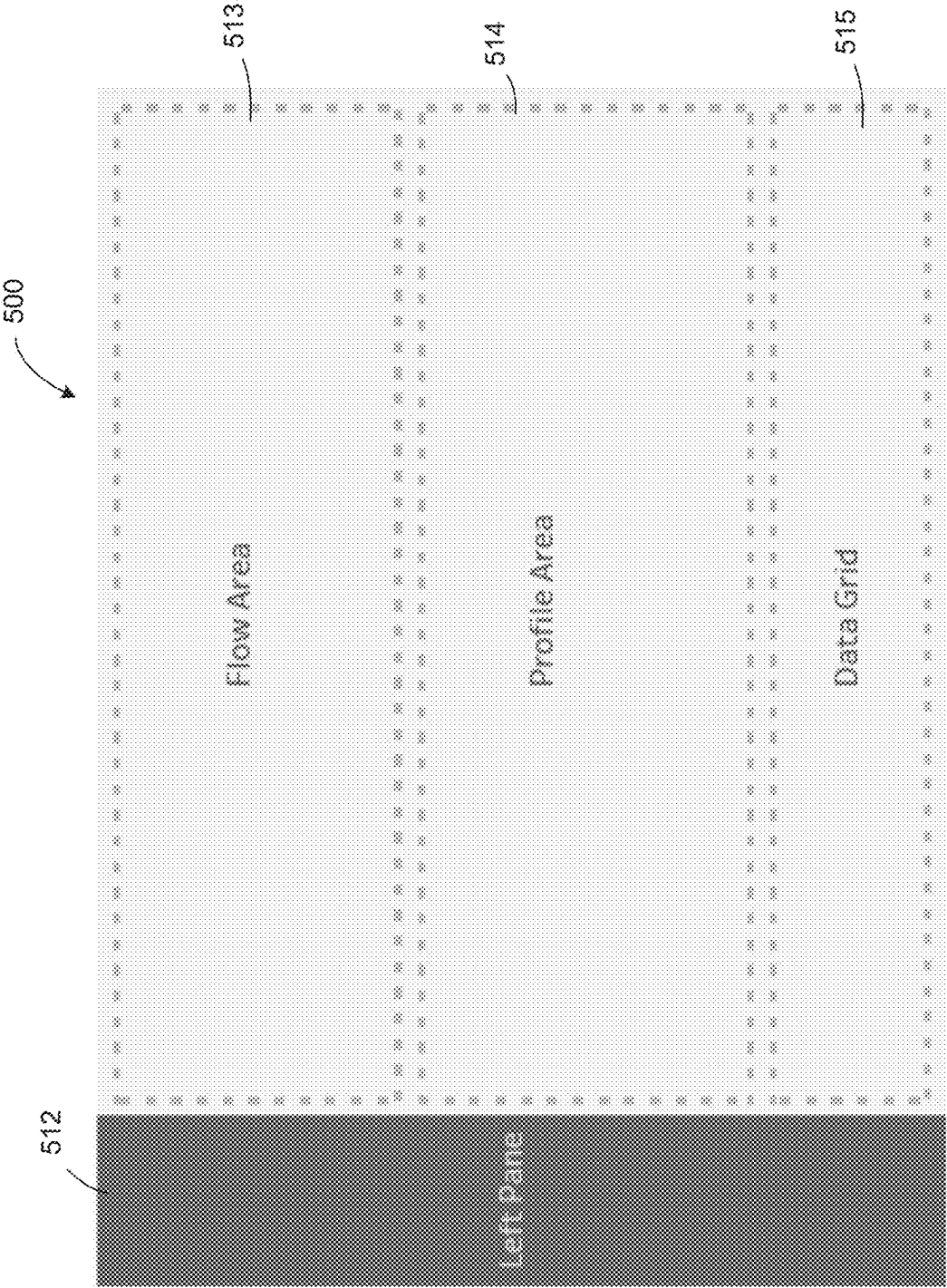
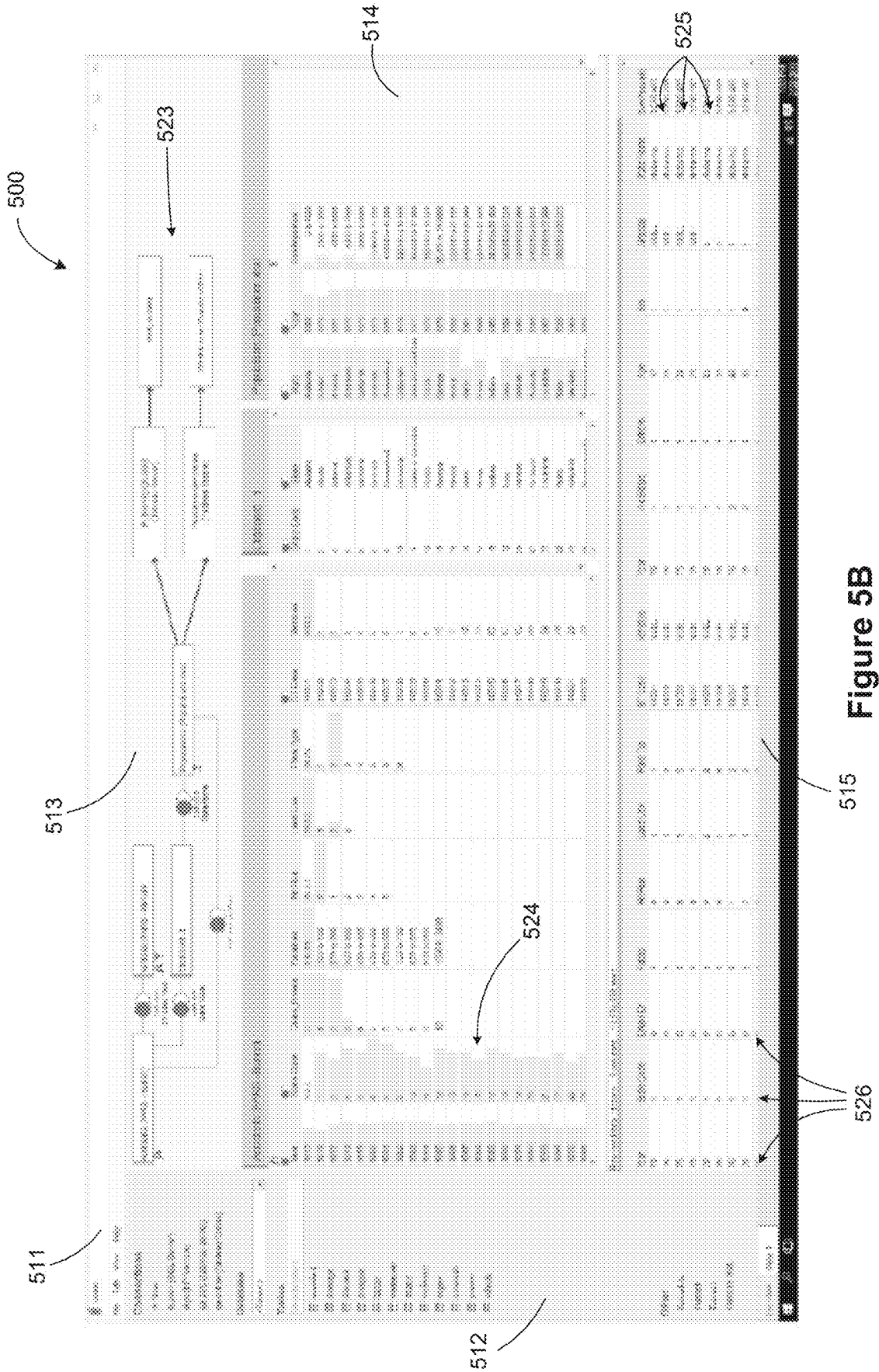


Figure 5A



600

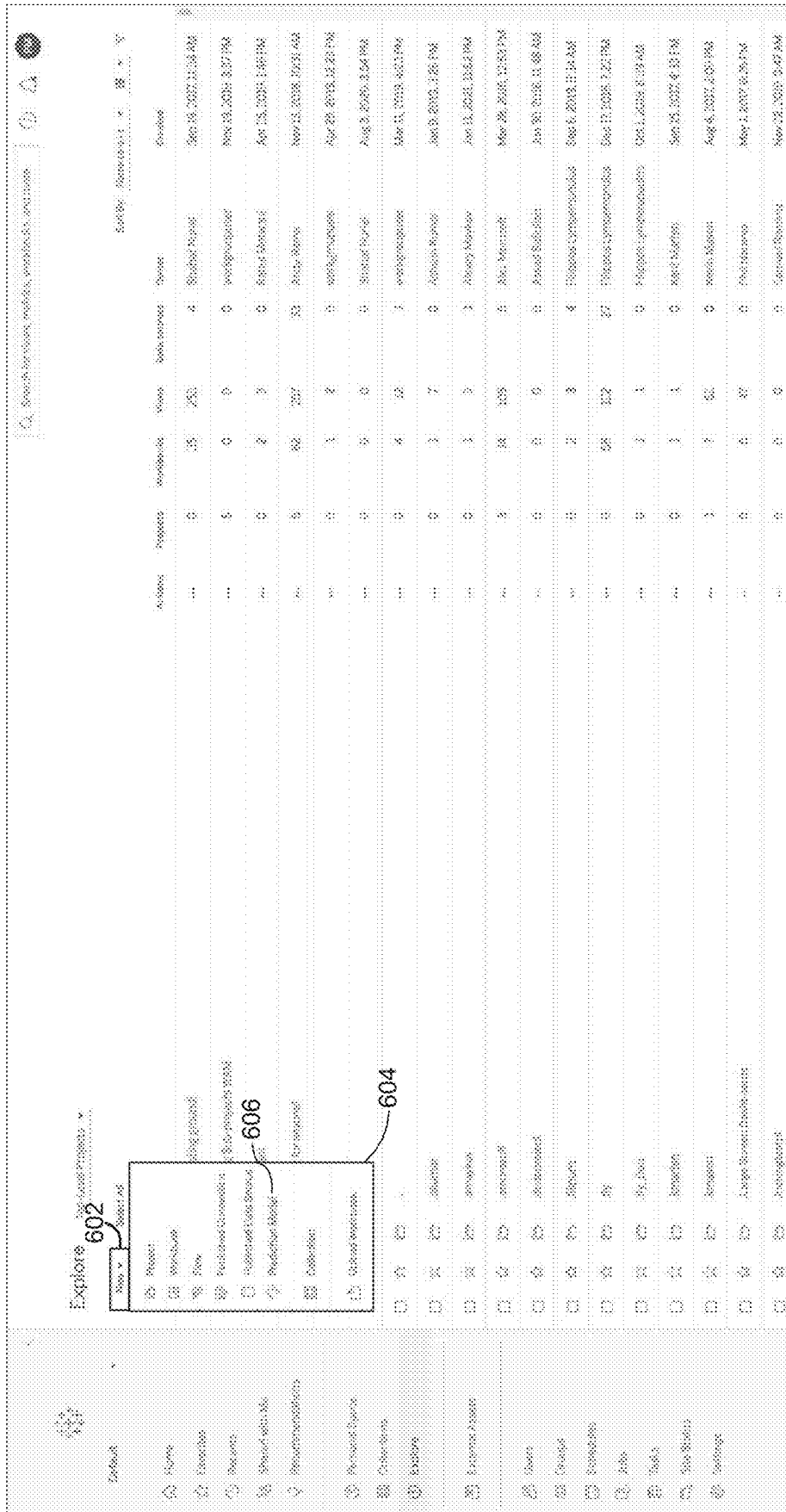


Figure 6A

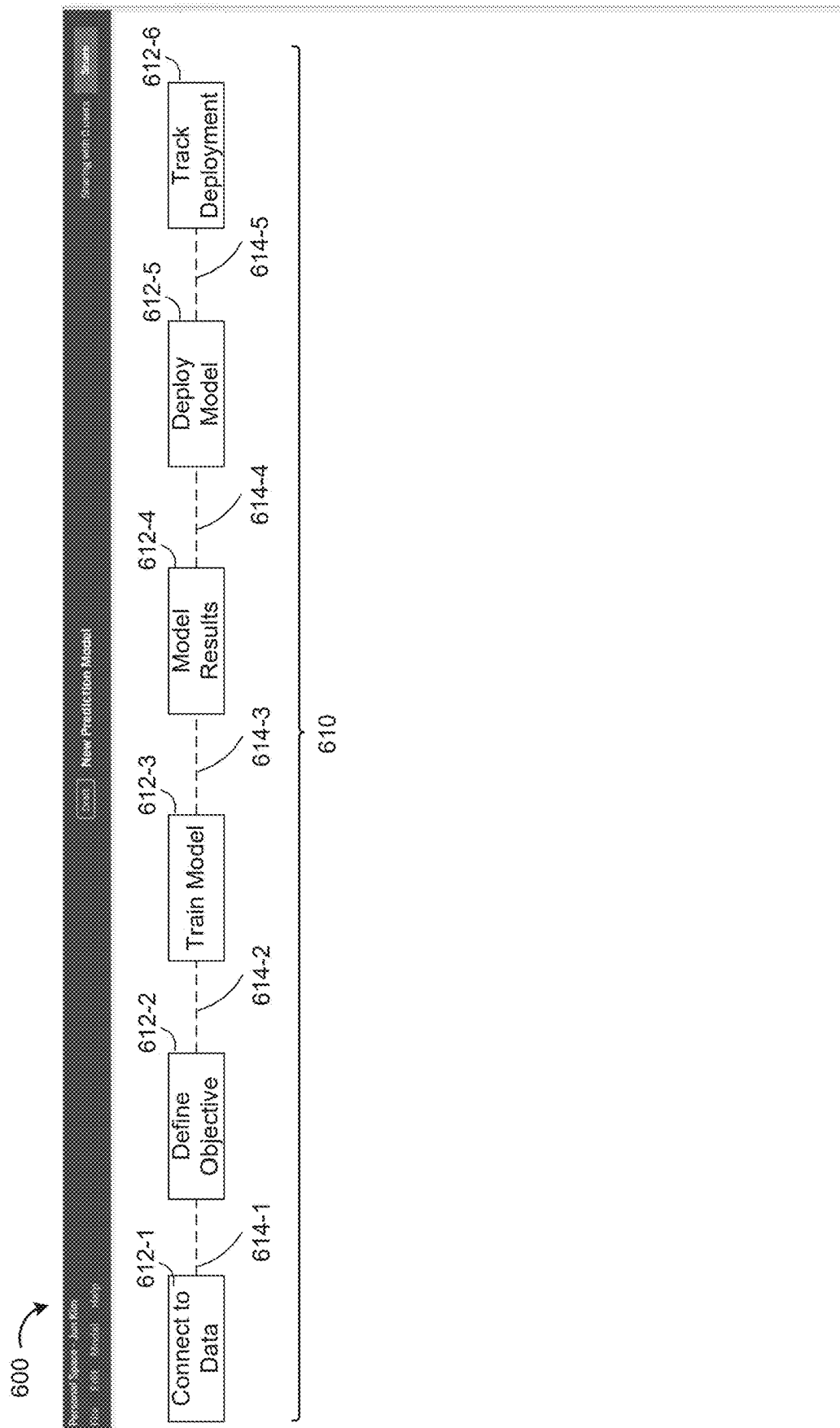


Figure 6B

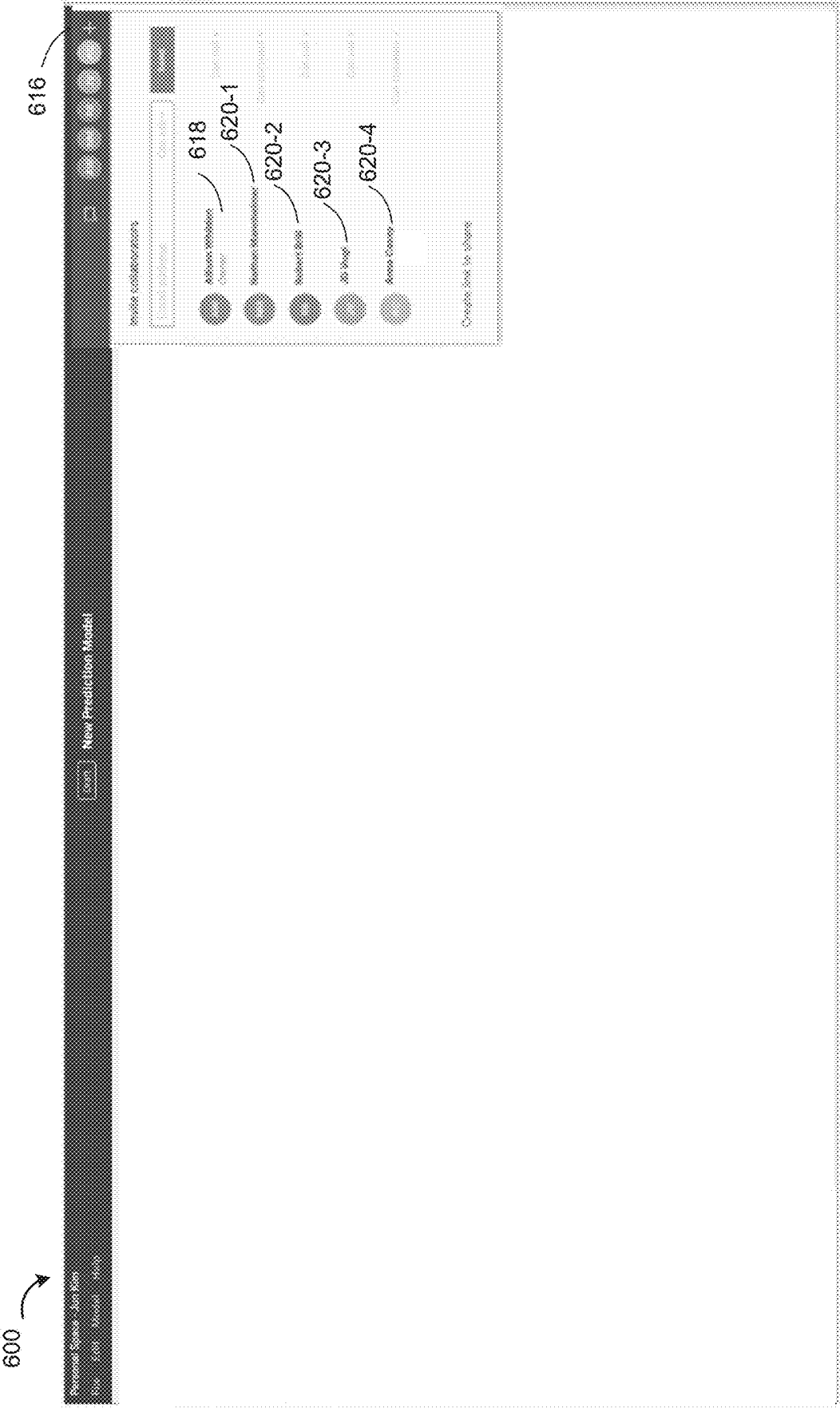


Figure 6C

600

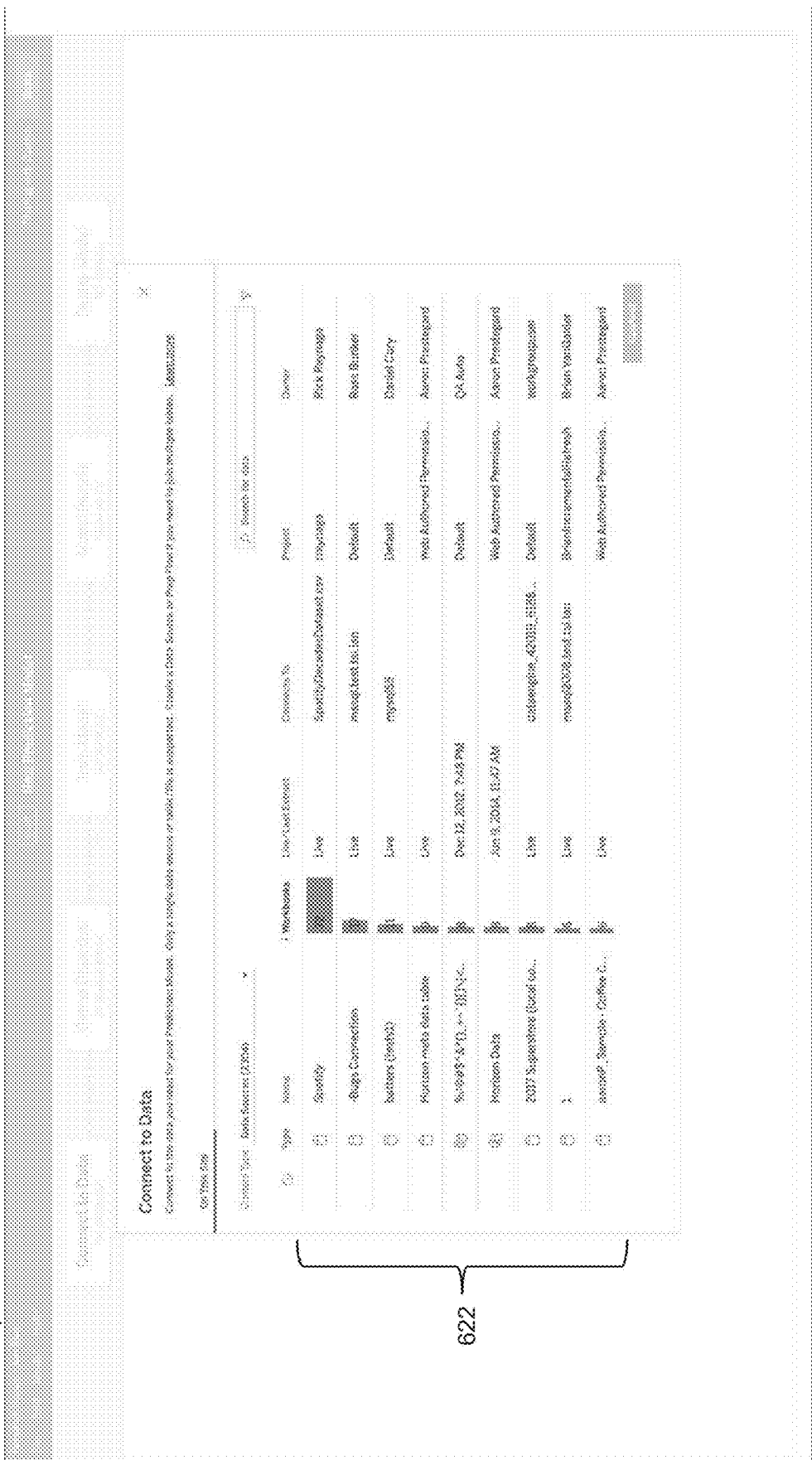


Figure 6D



Figure 6E

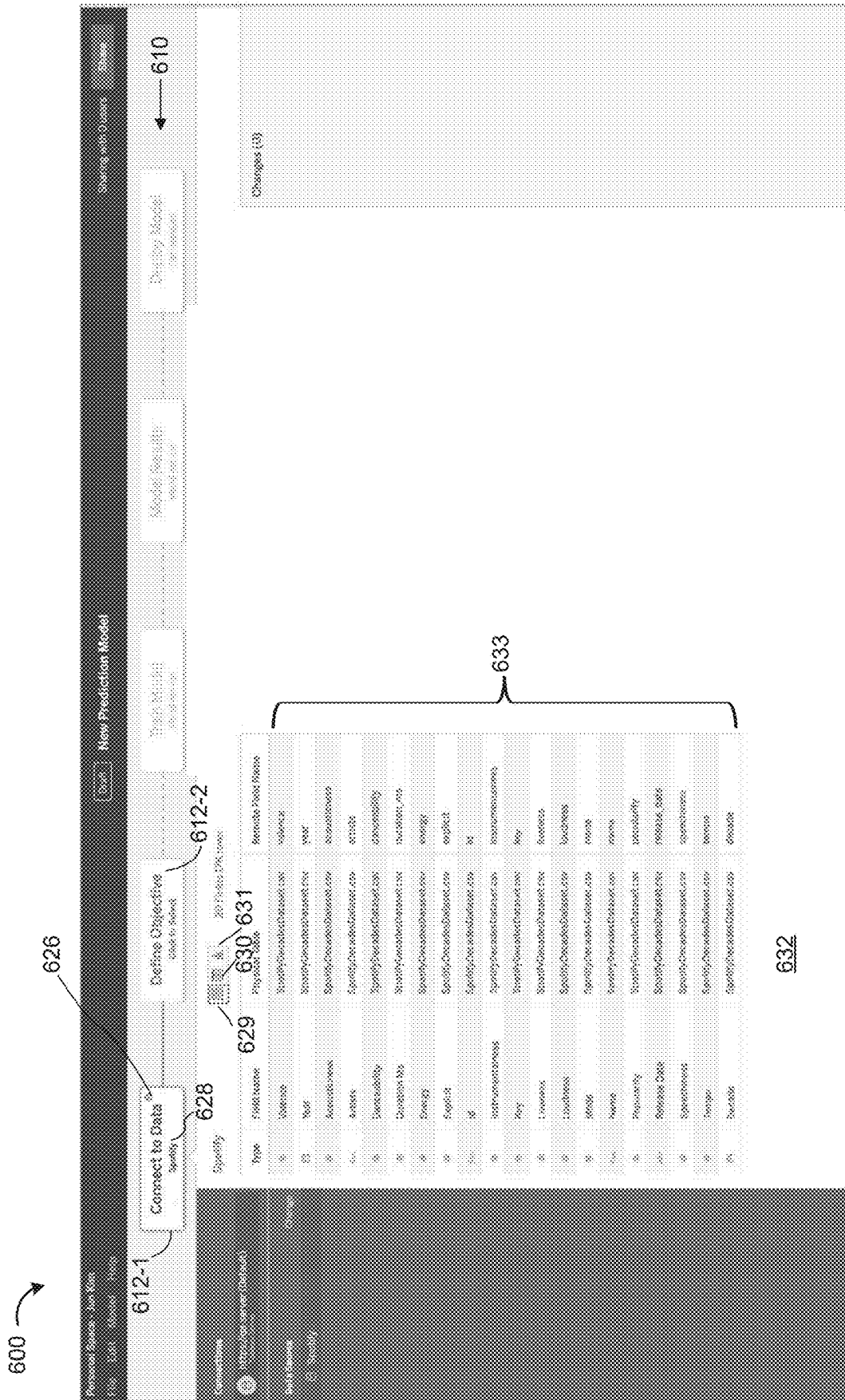
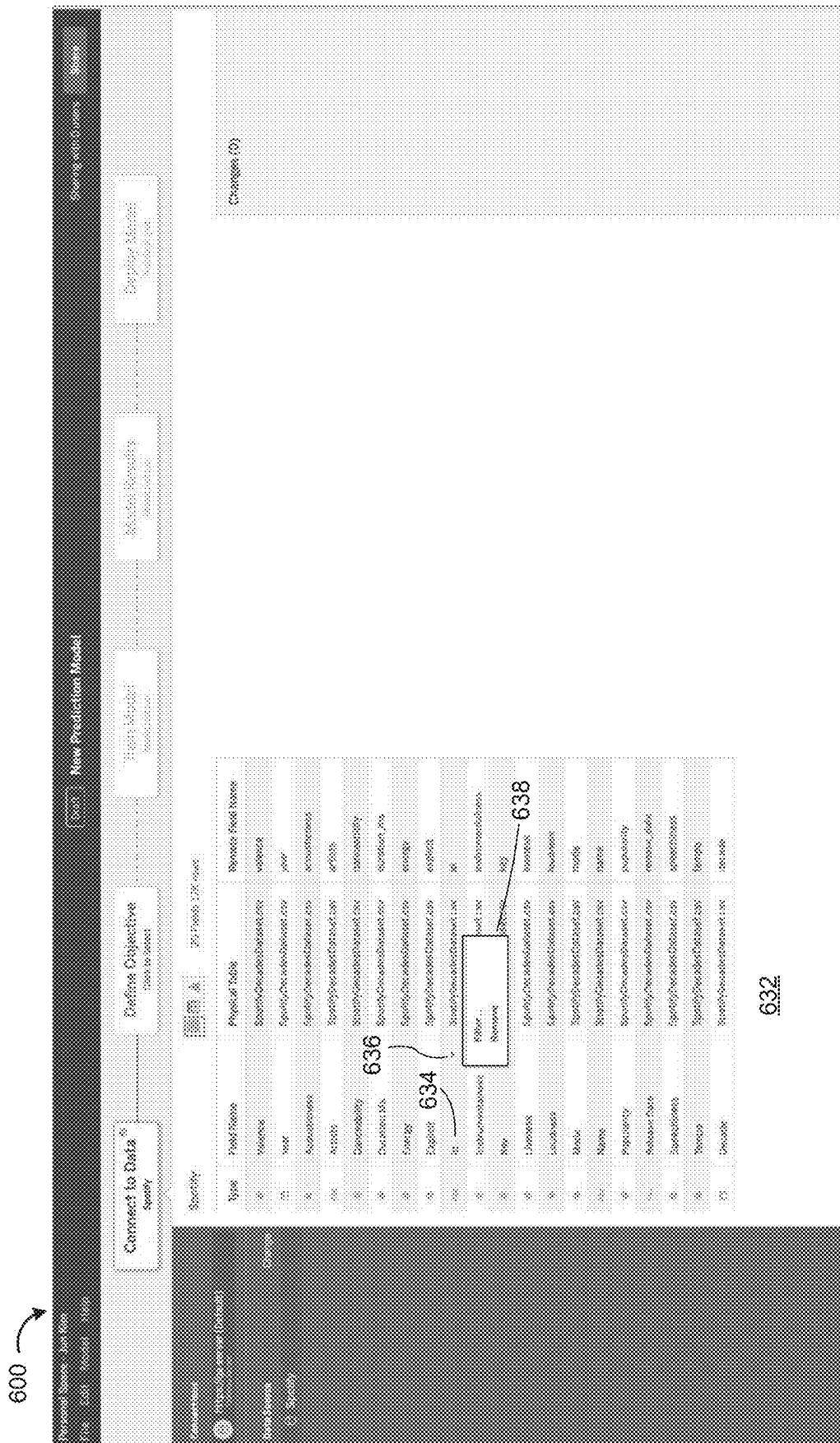
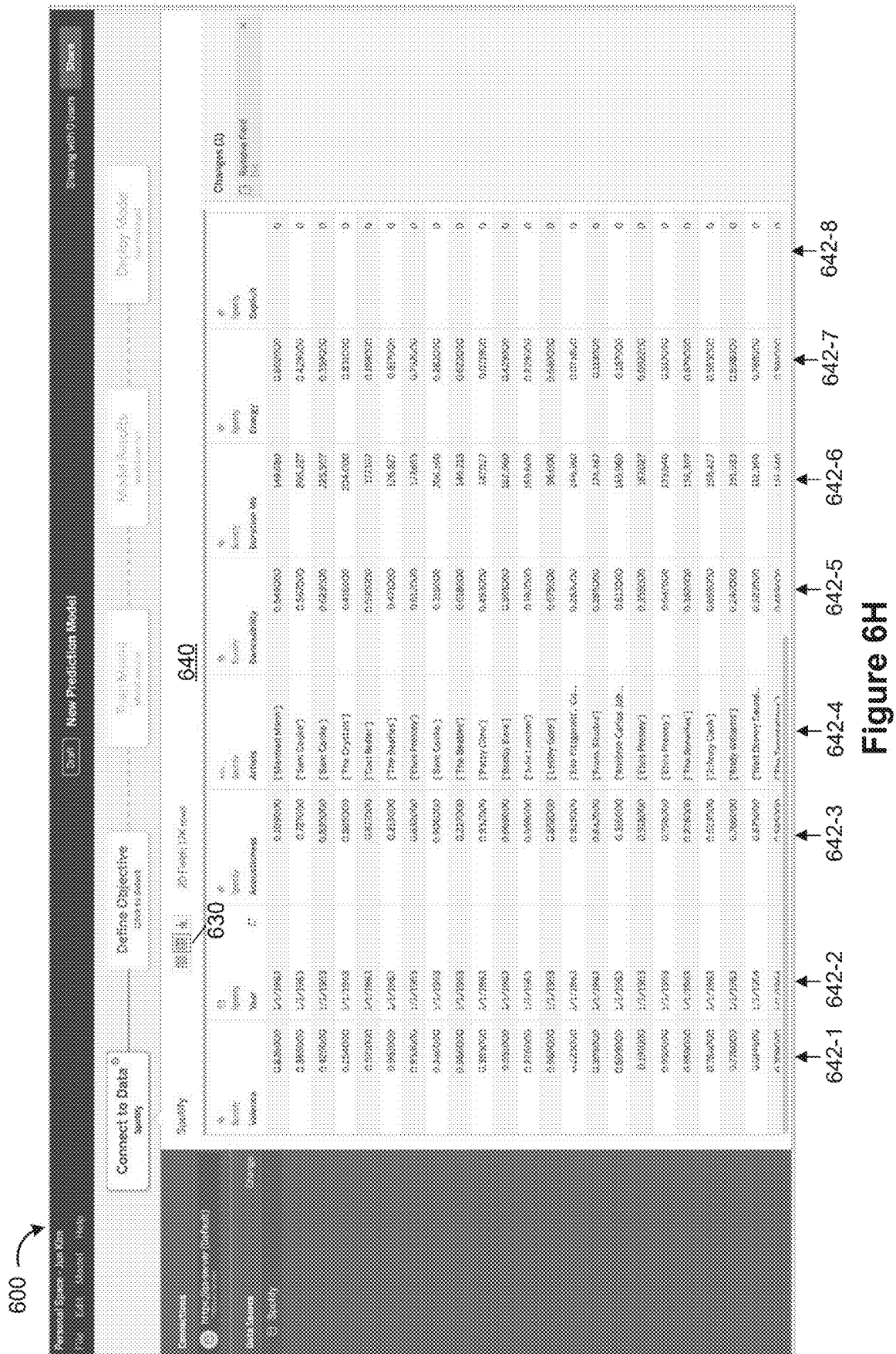


Figure 6F







600

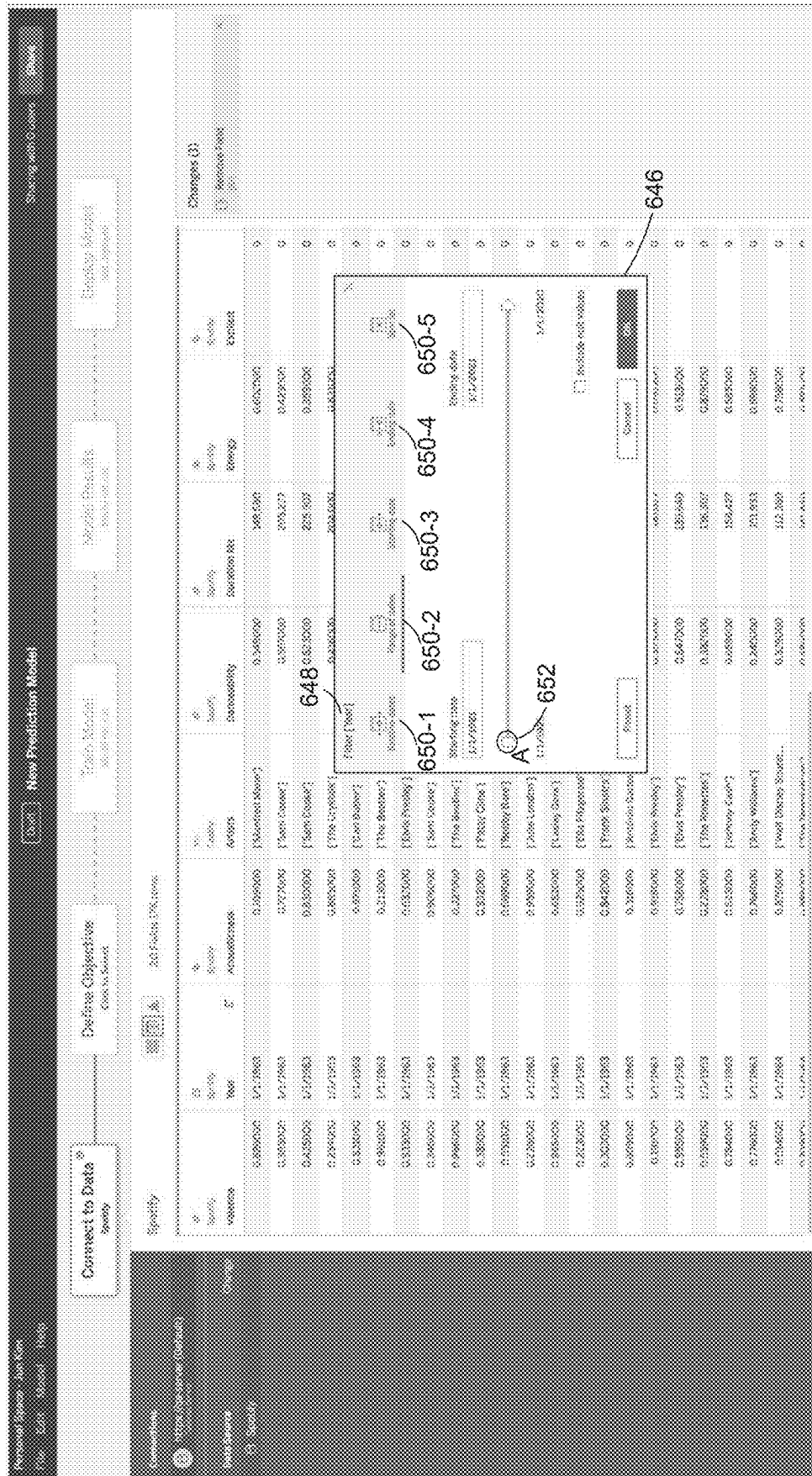
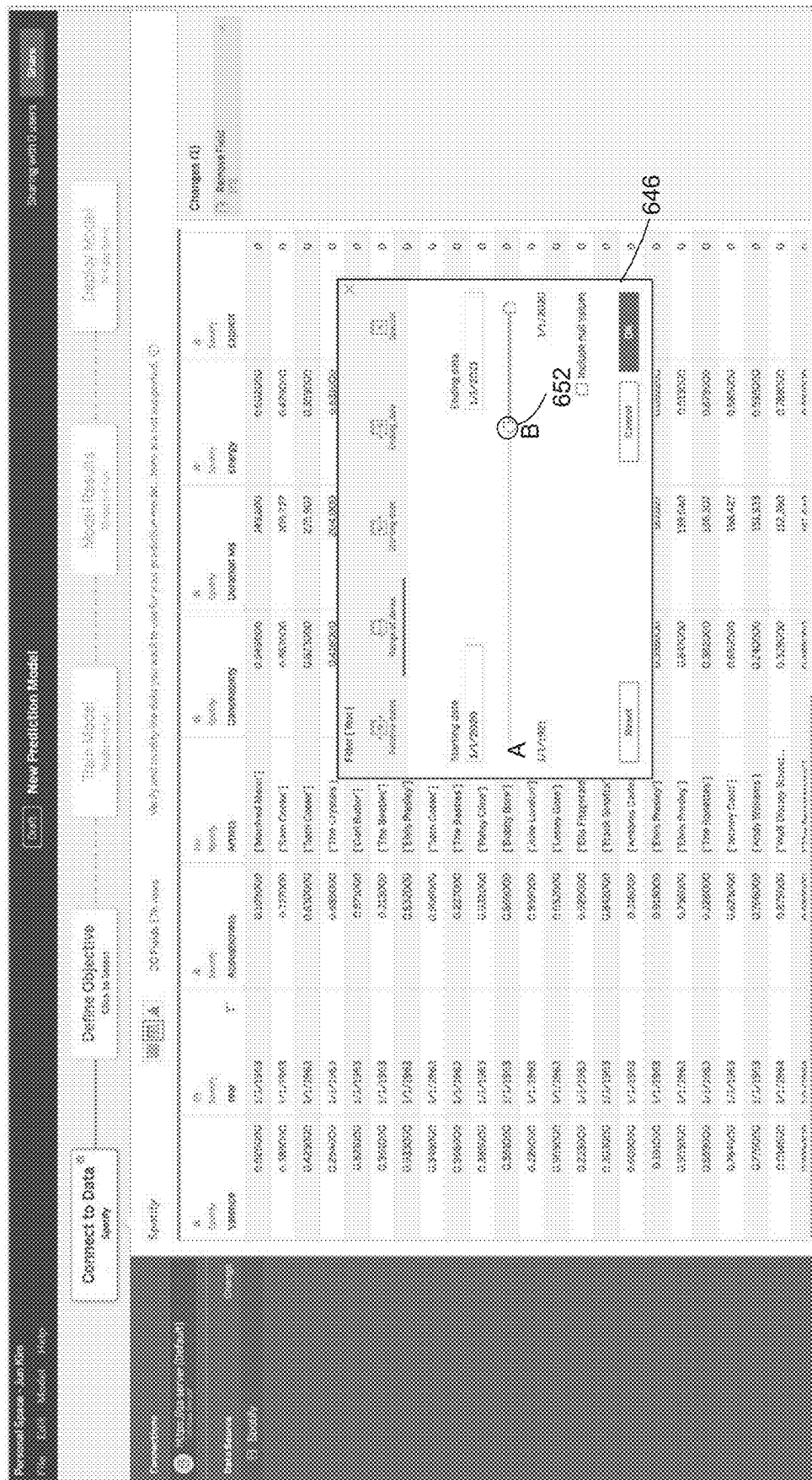


Figure 6J



600

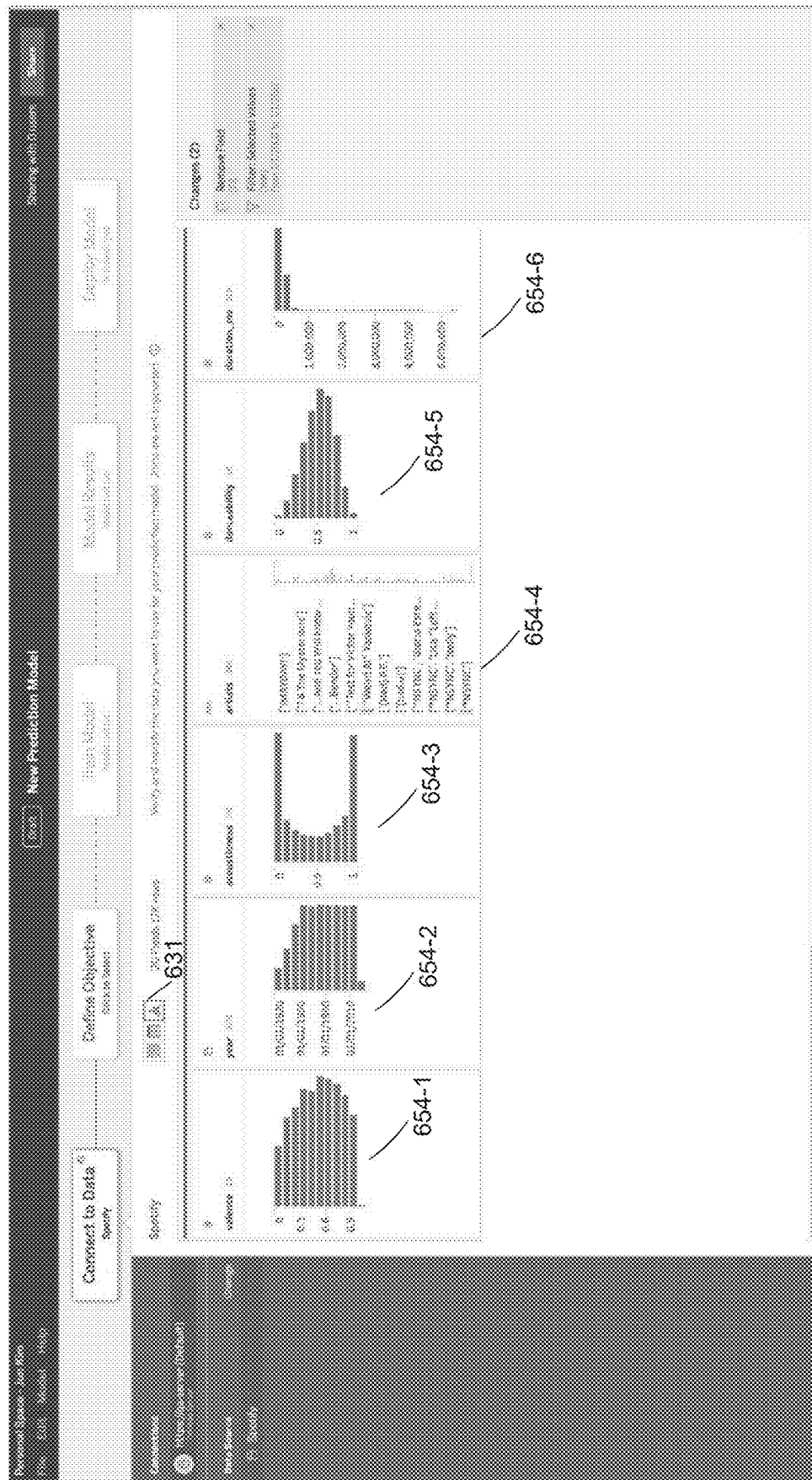


Figure 6L

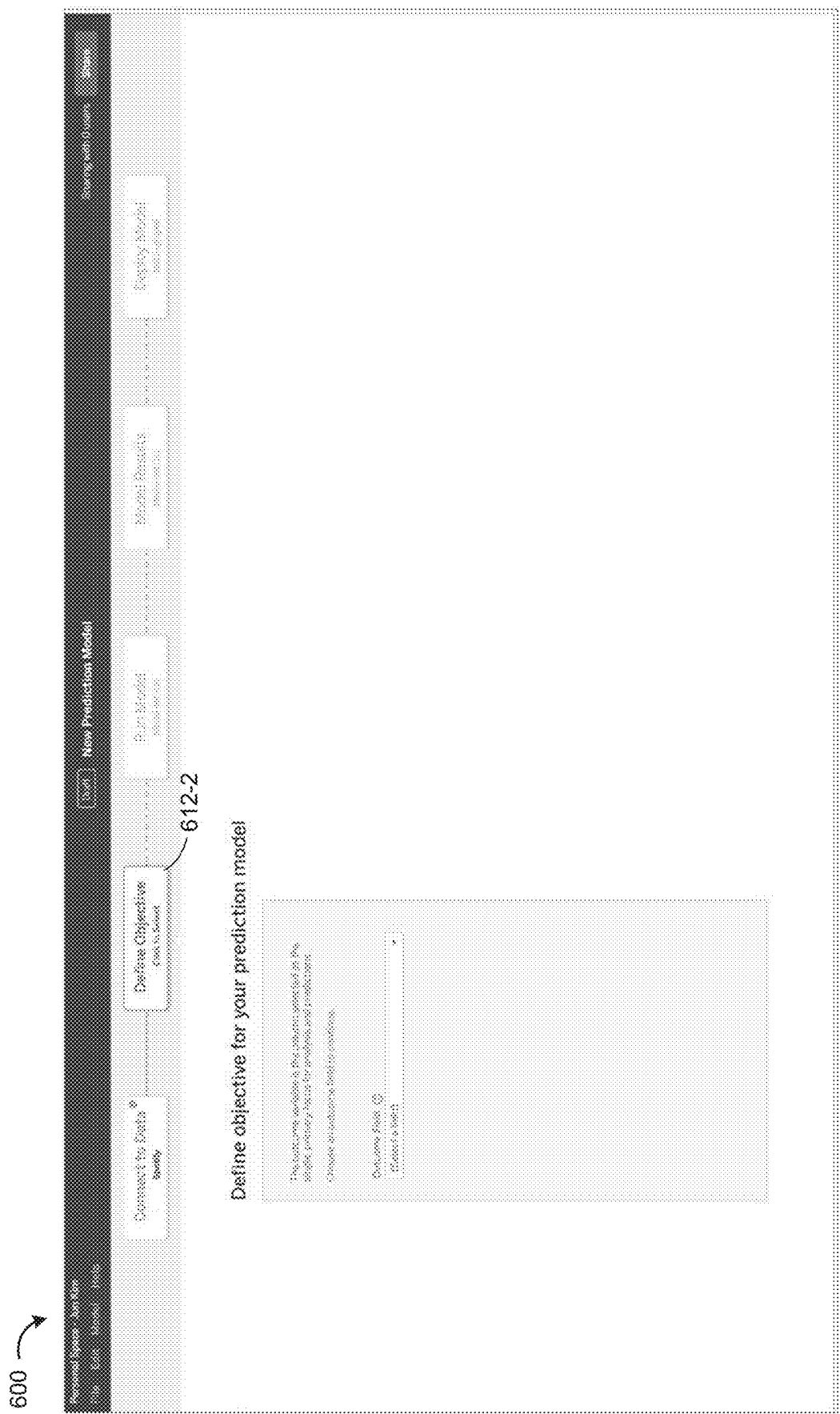


Figure 6M

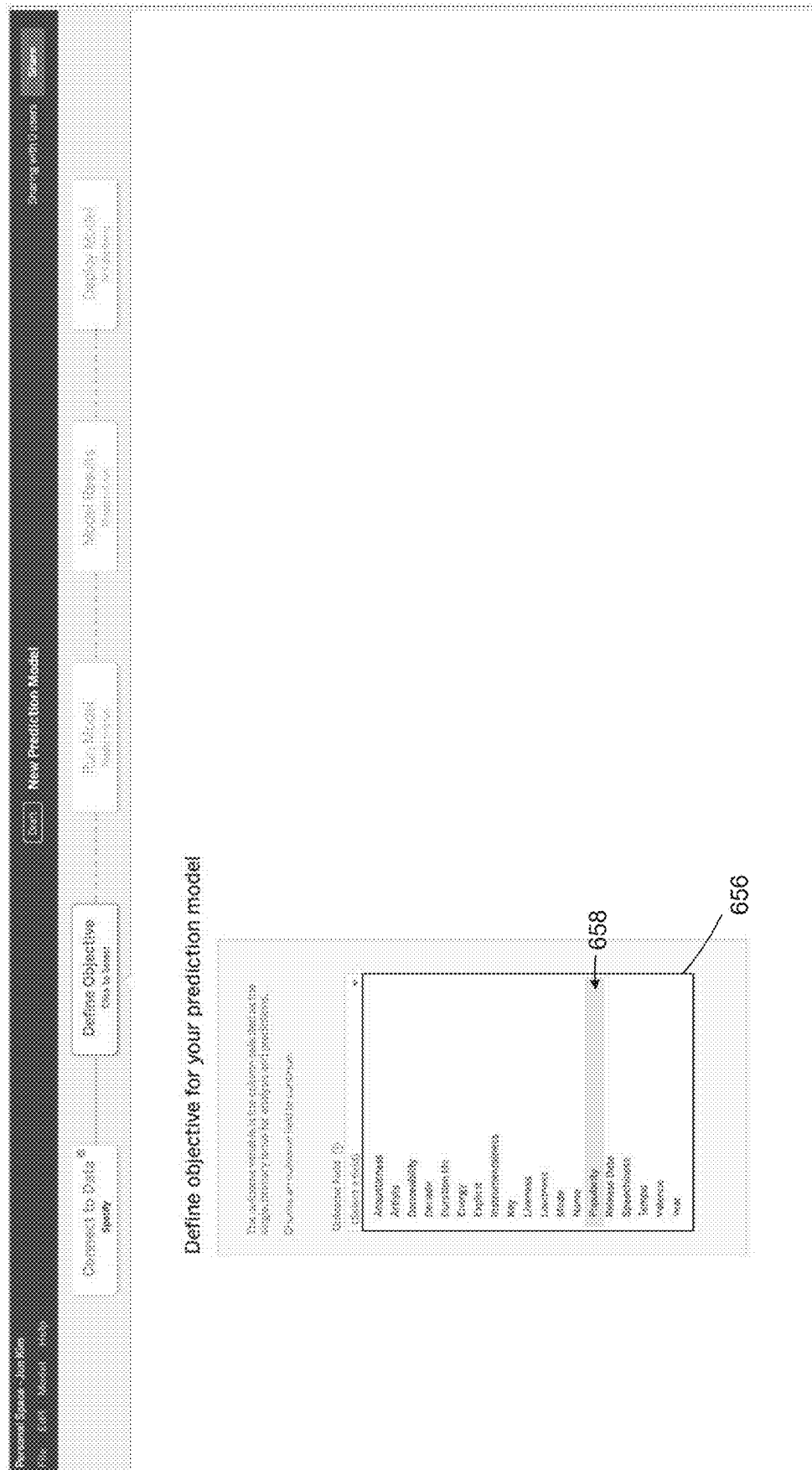


Figure 62

600

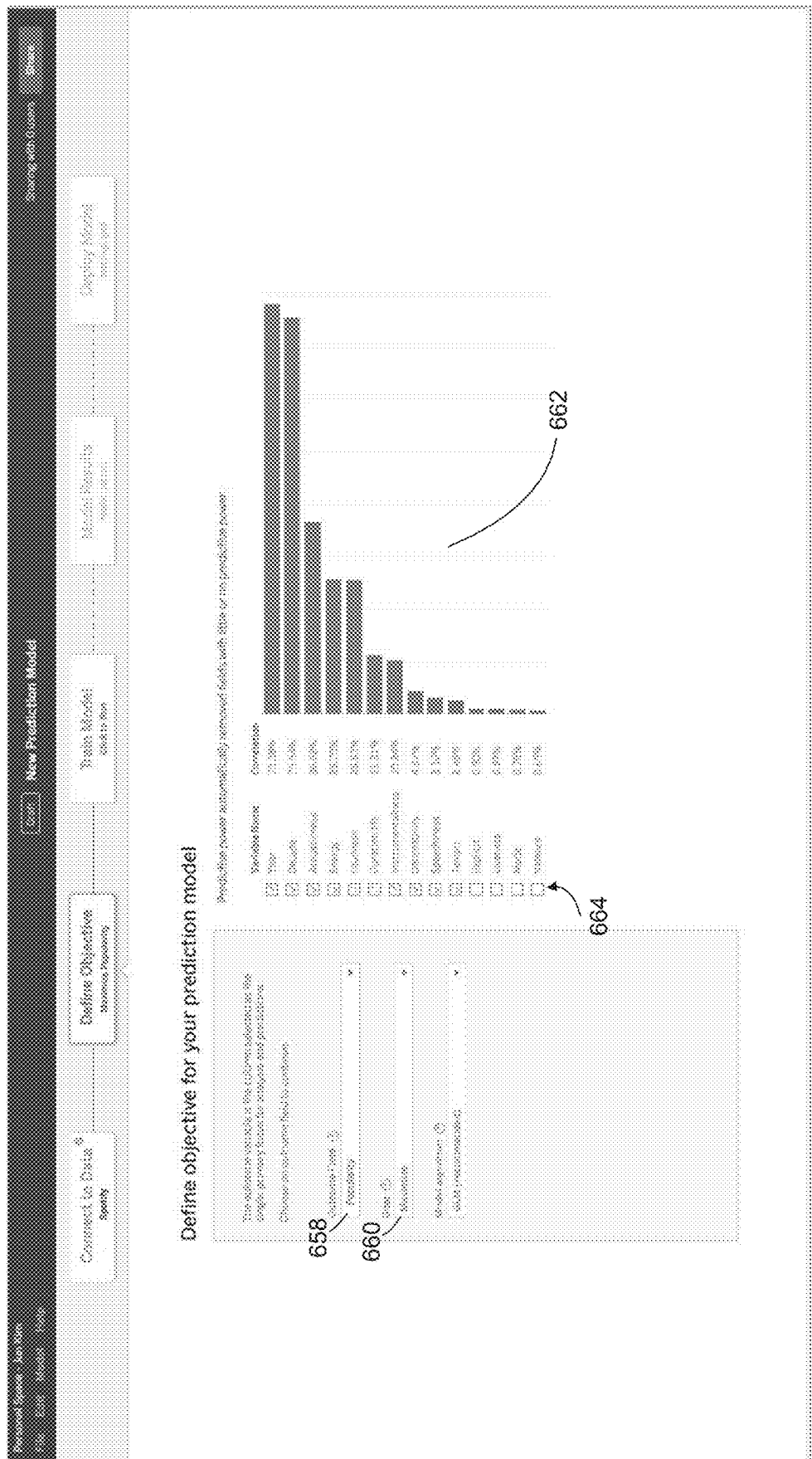


Figure 60

600

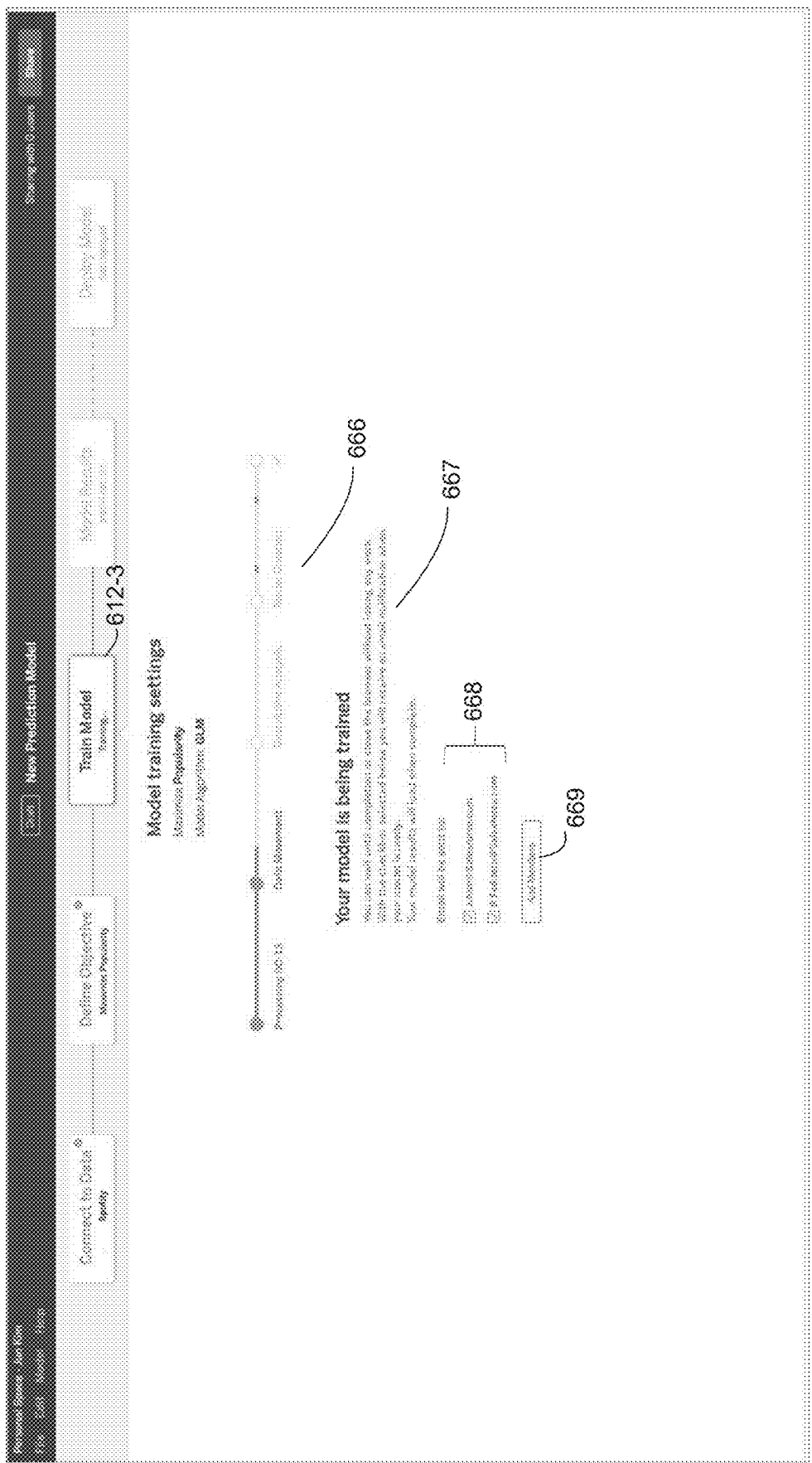
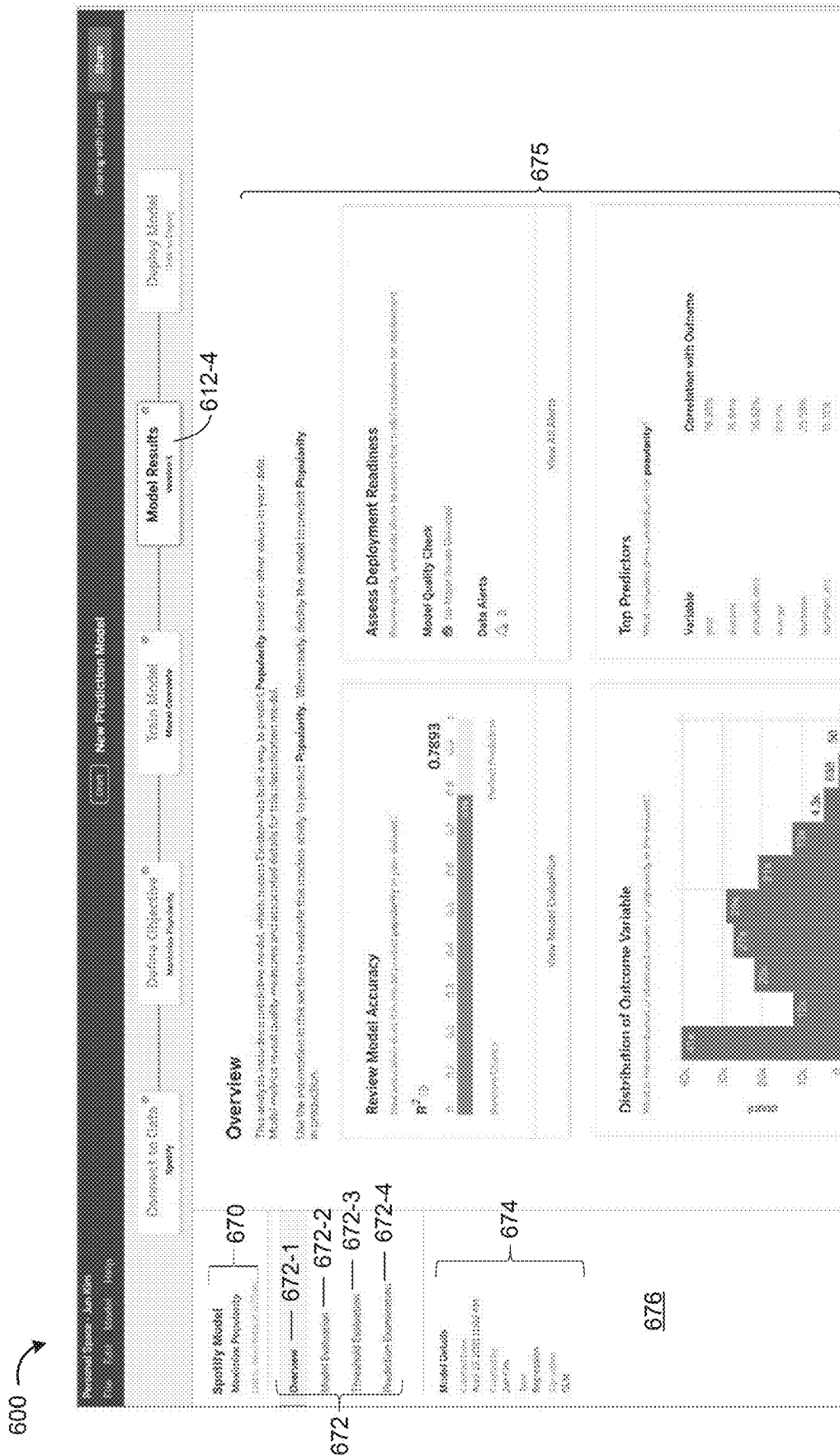


Figure 6P



600



Figure 6R



Figure 6S

600



Figure 6T

600

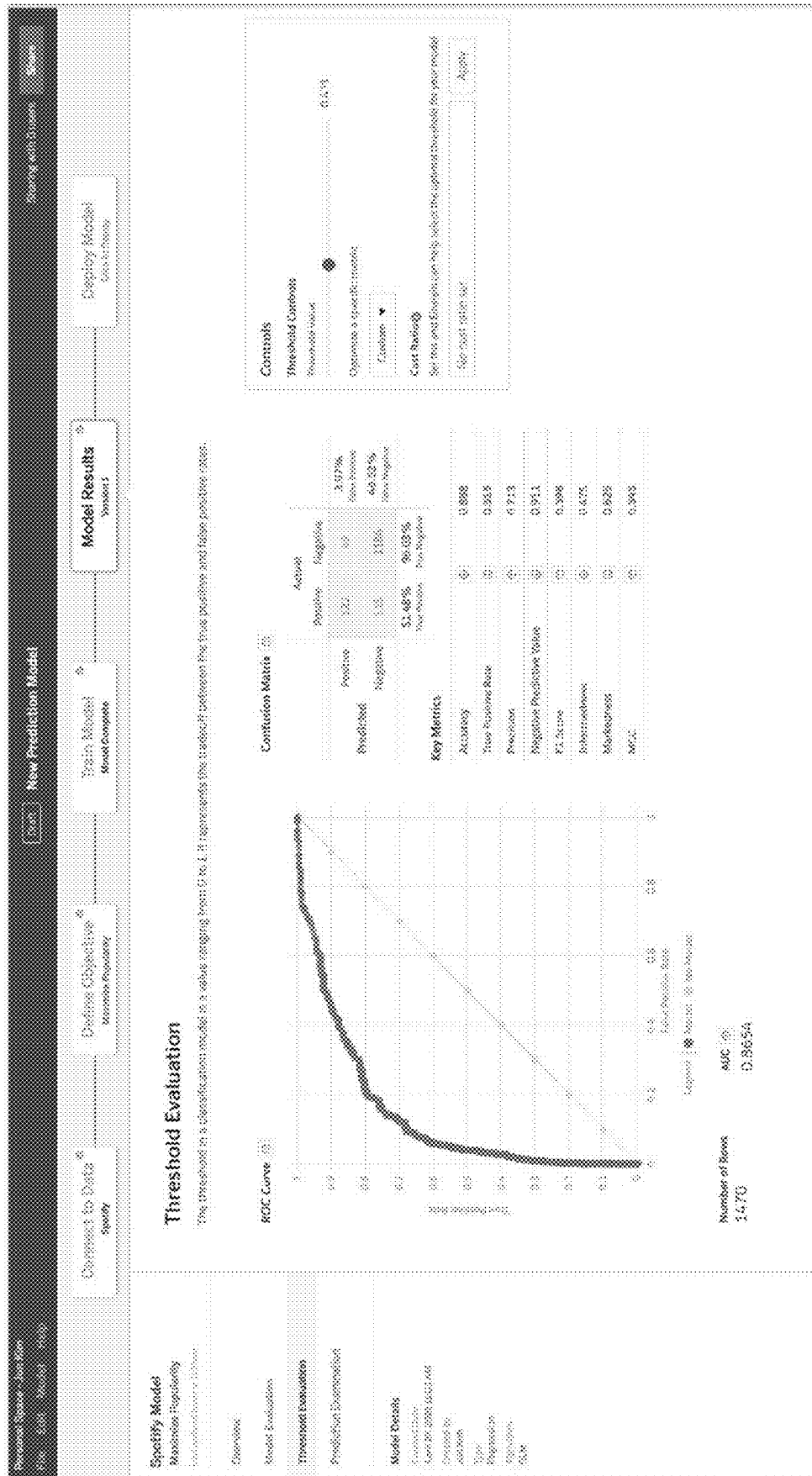


Figure 6U

000



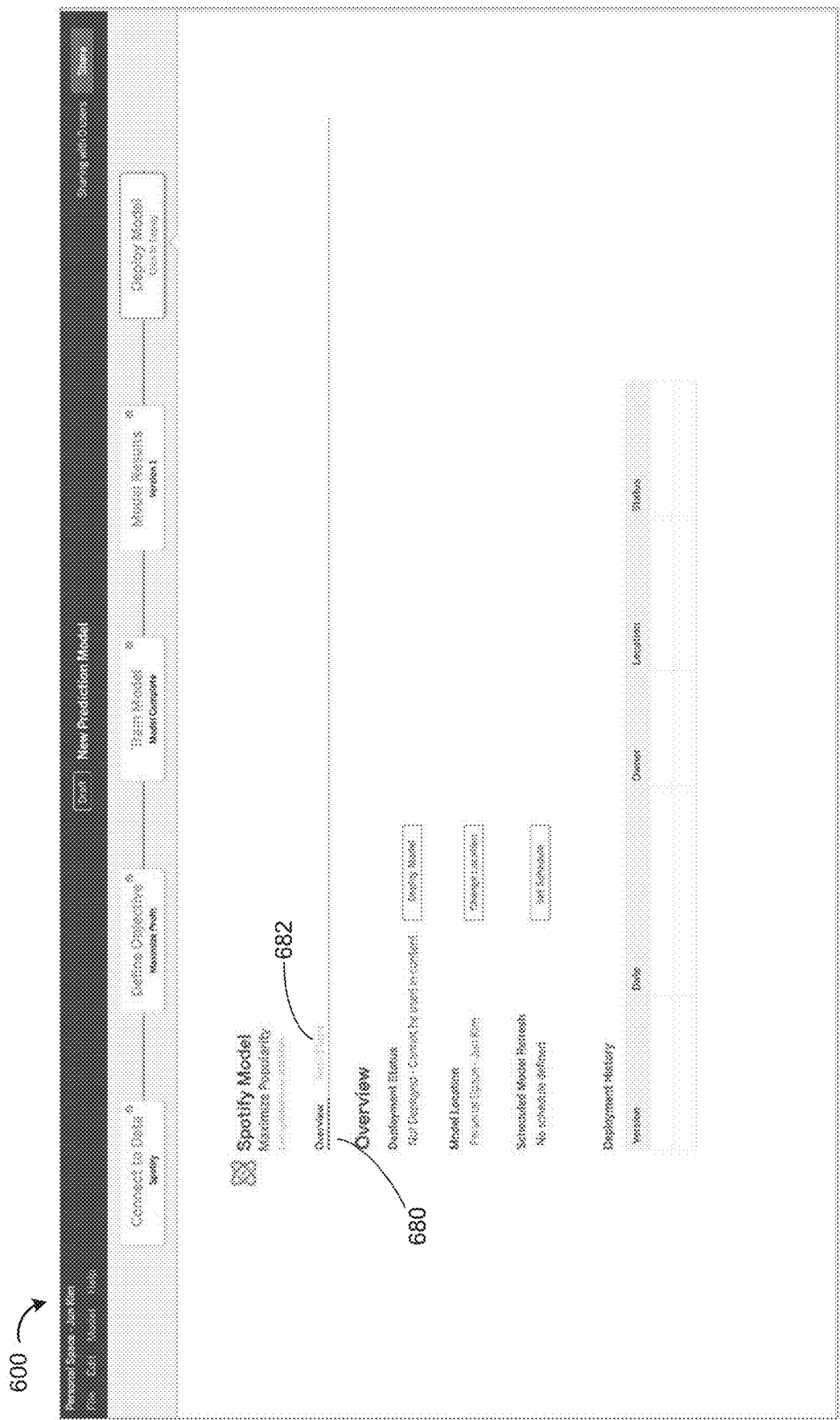


Figure 6W

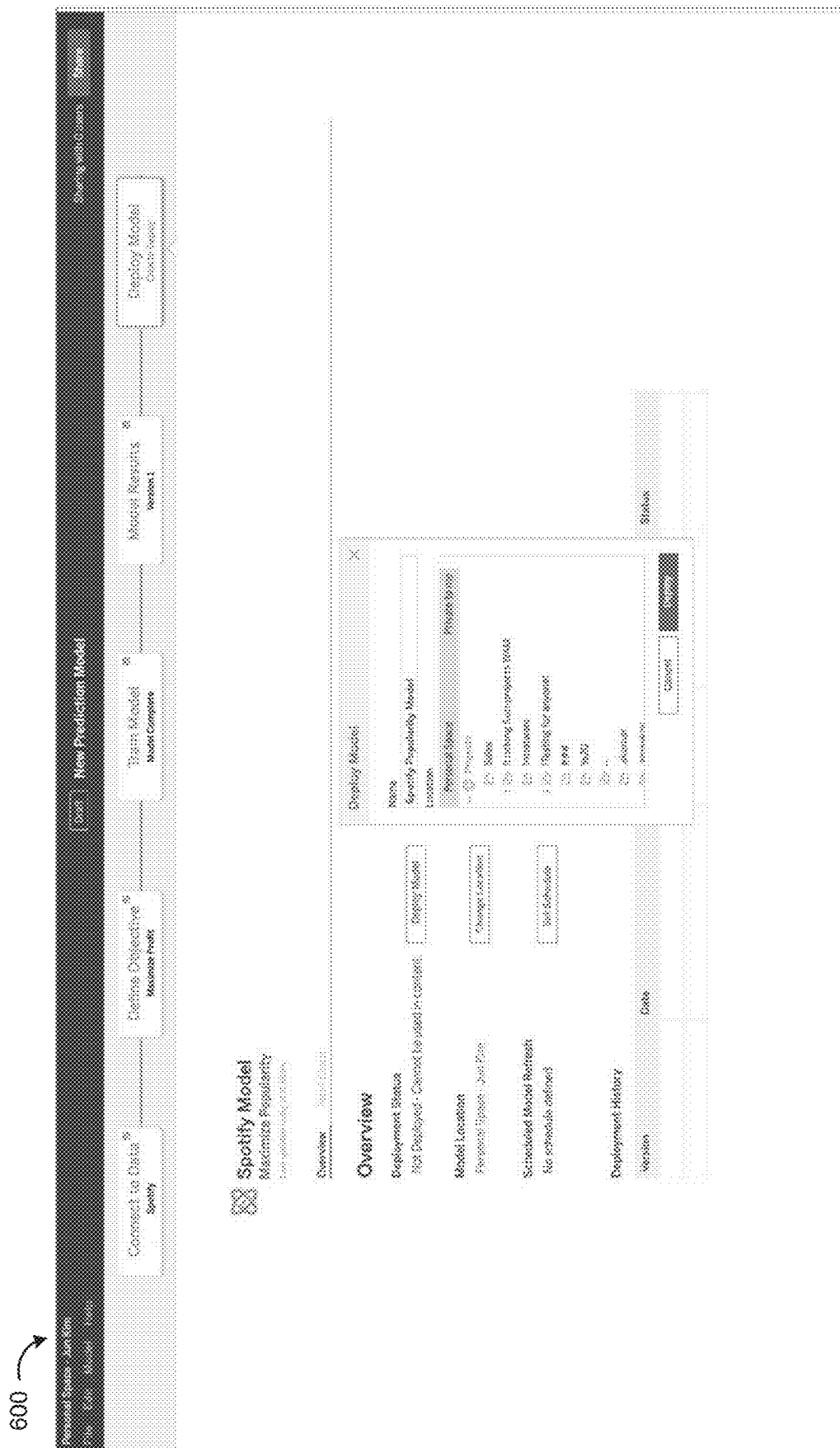


Figure 6X

600

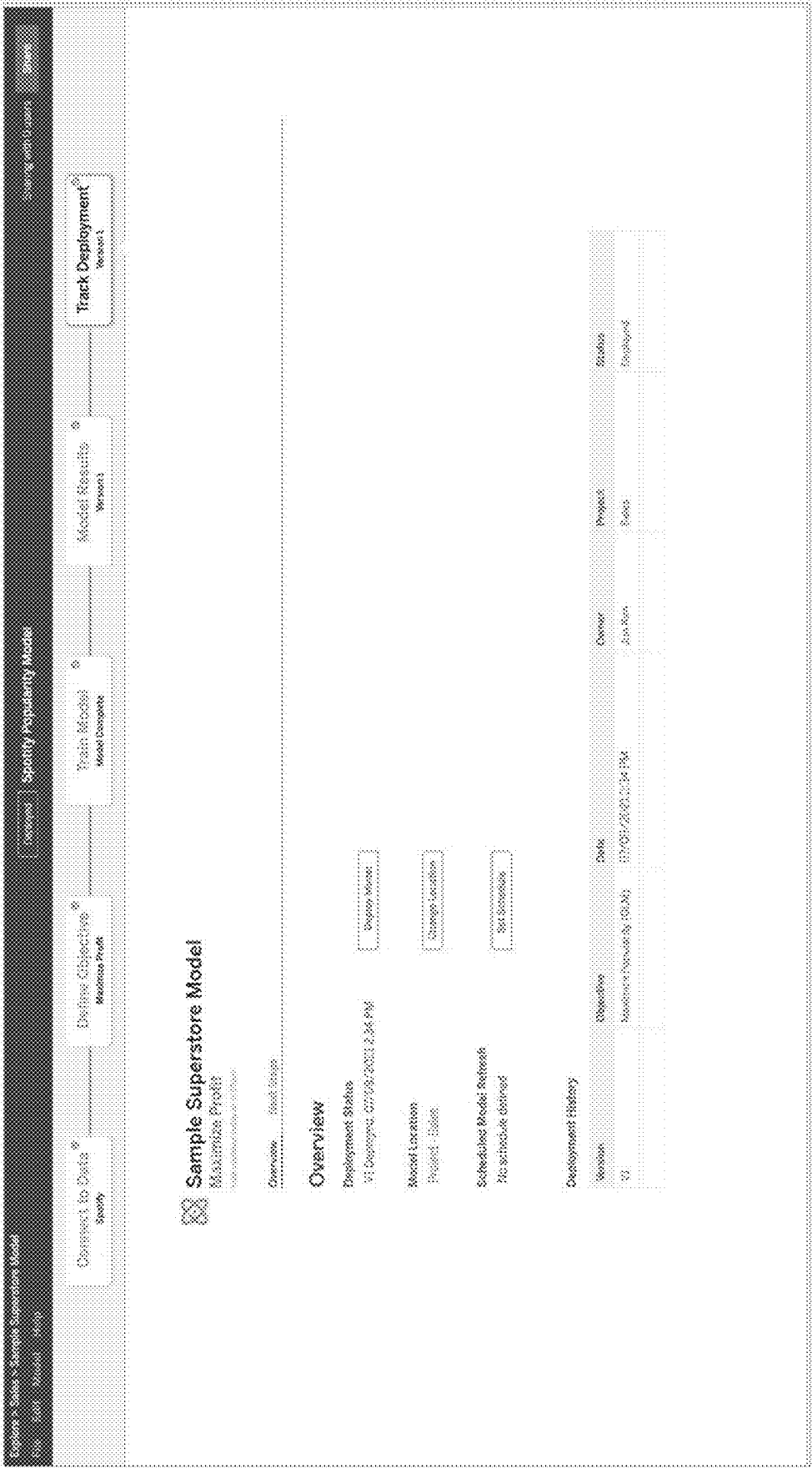


Figure 6Y

600

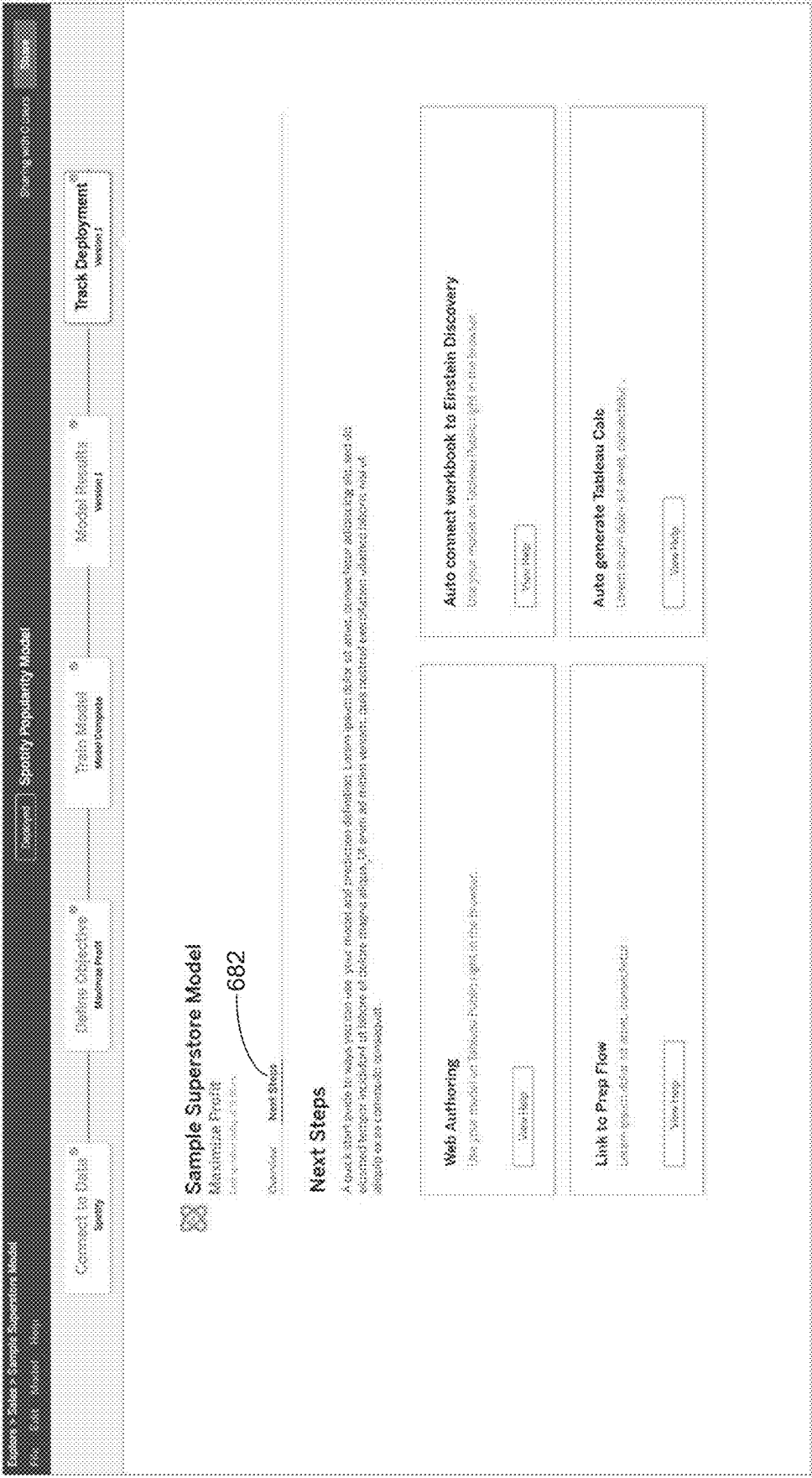


Figure 6Z



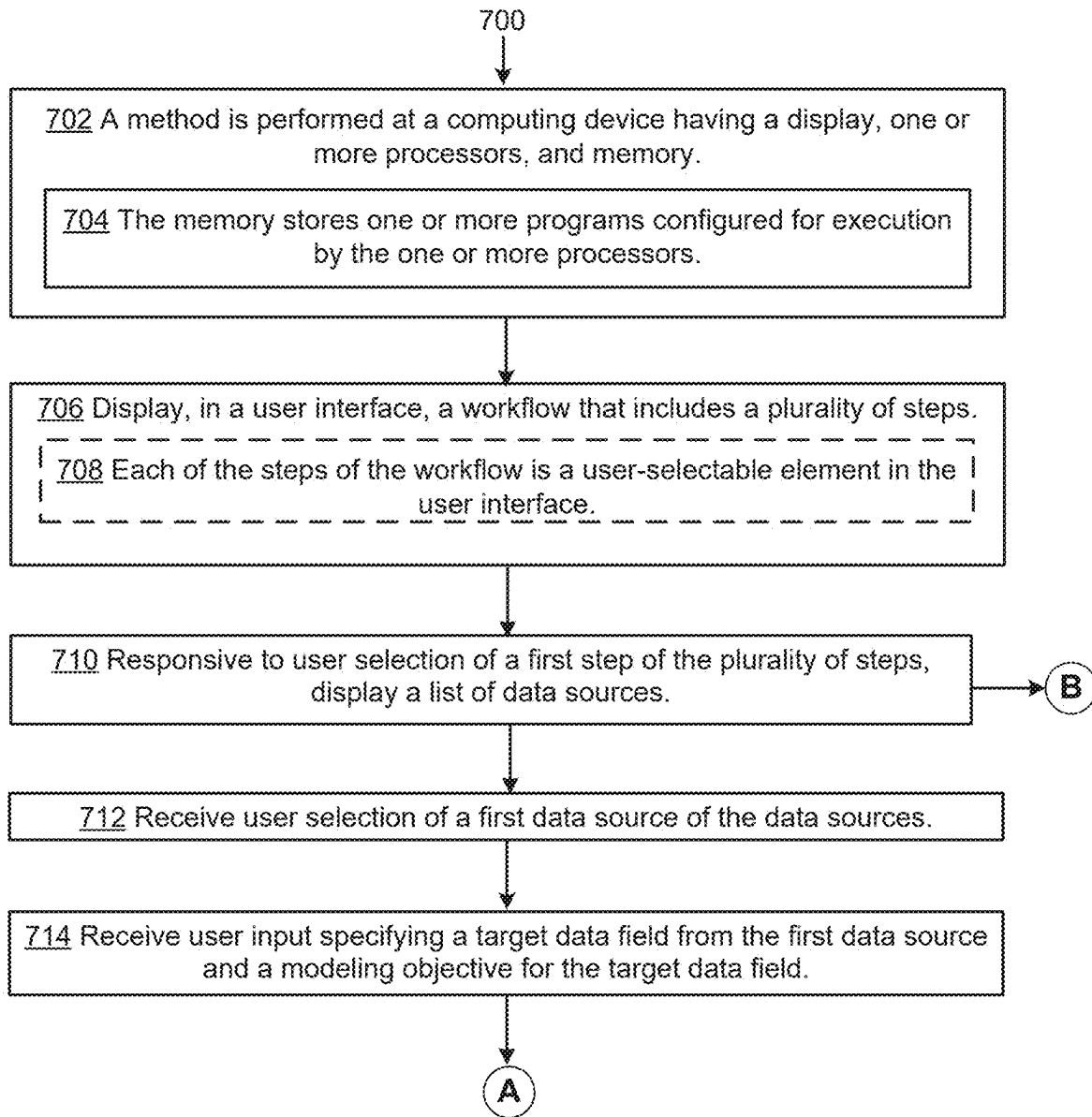
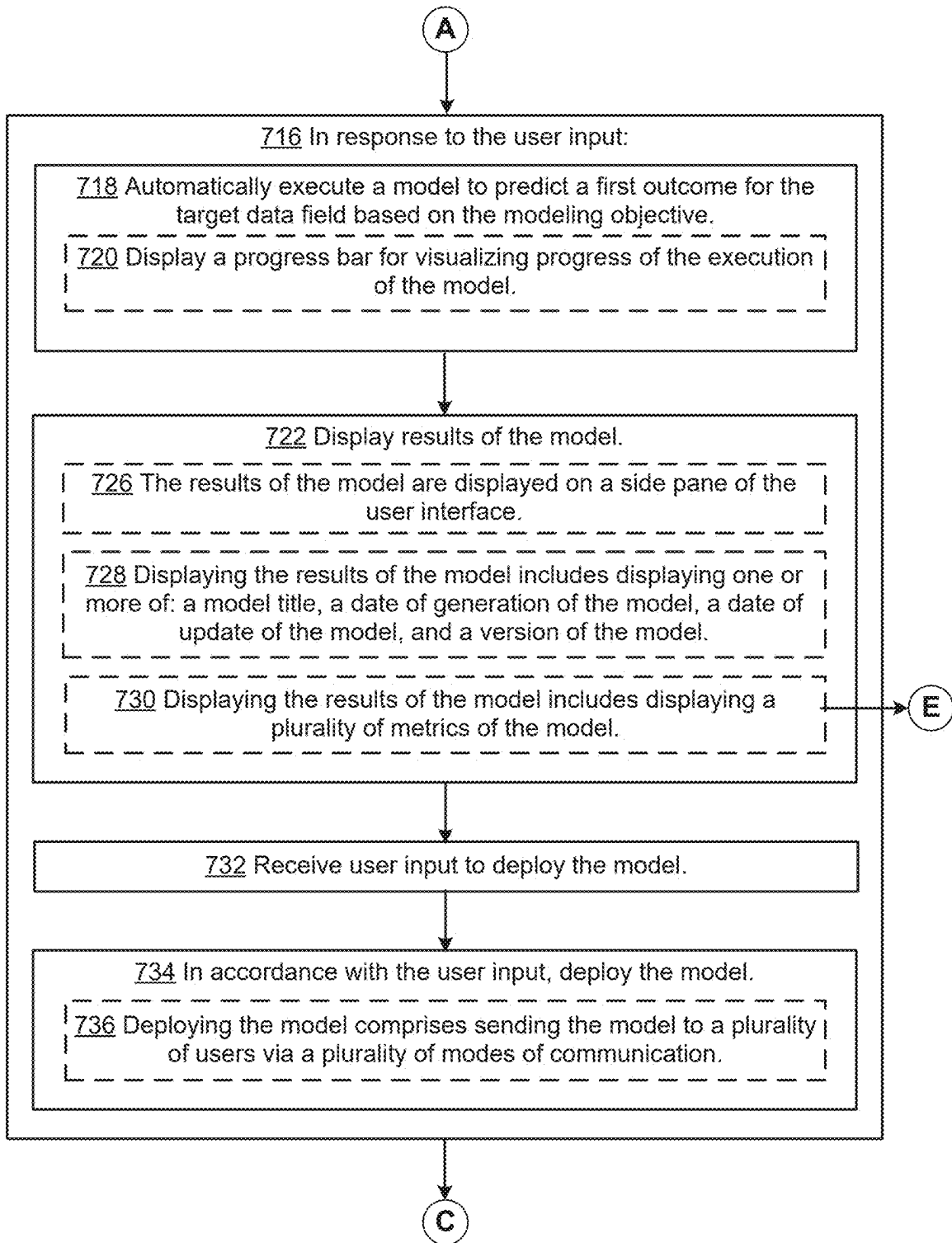


Figure 7A

**Figure 7B**

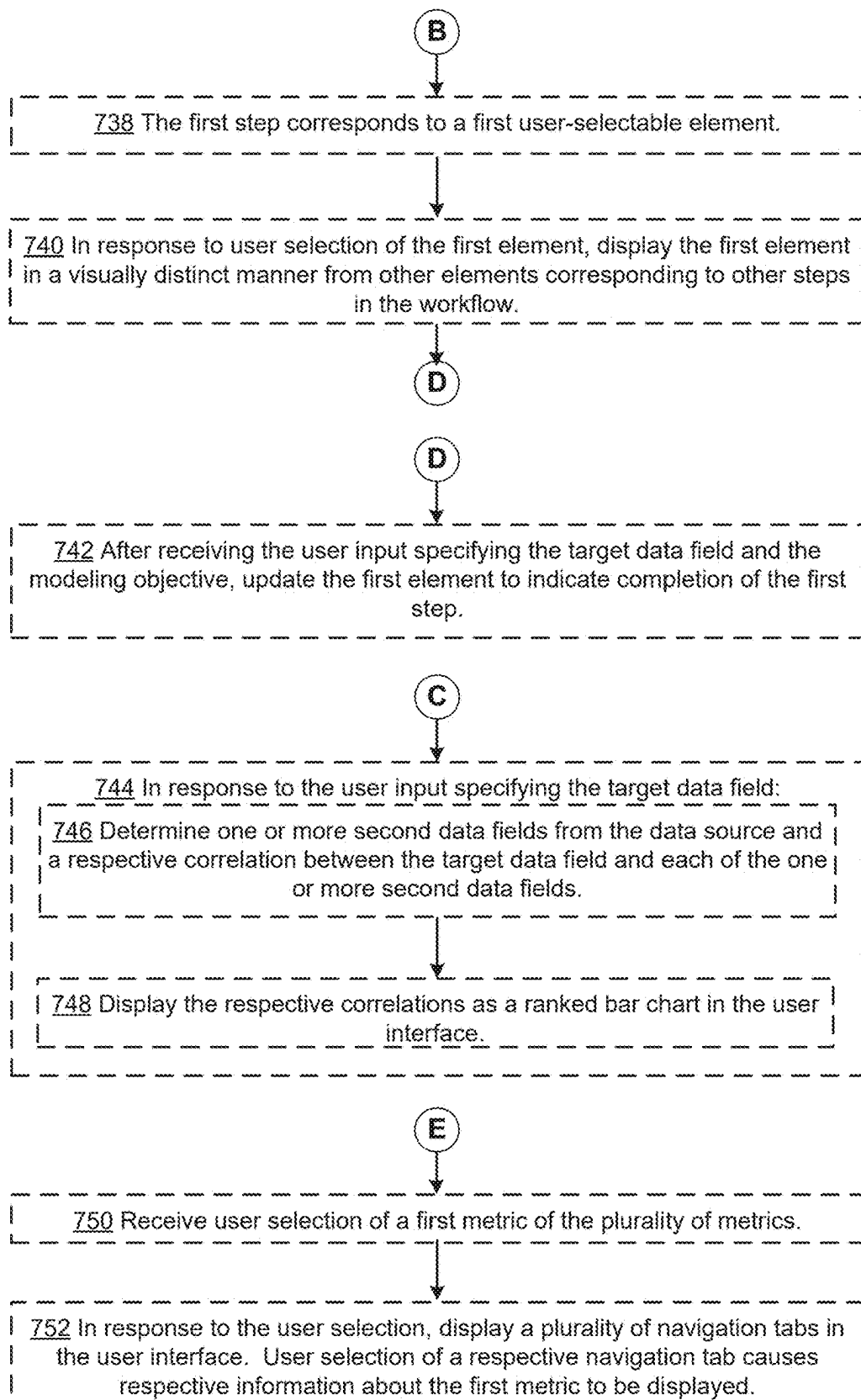


Figure 7C

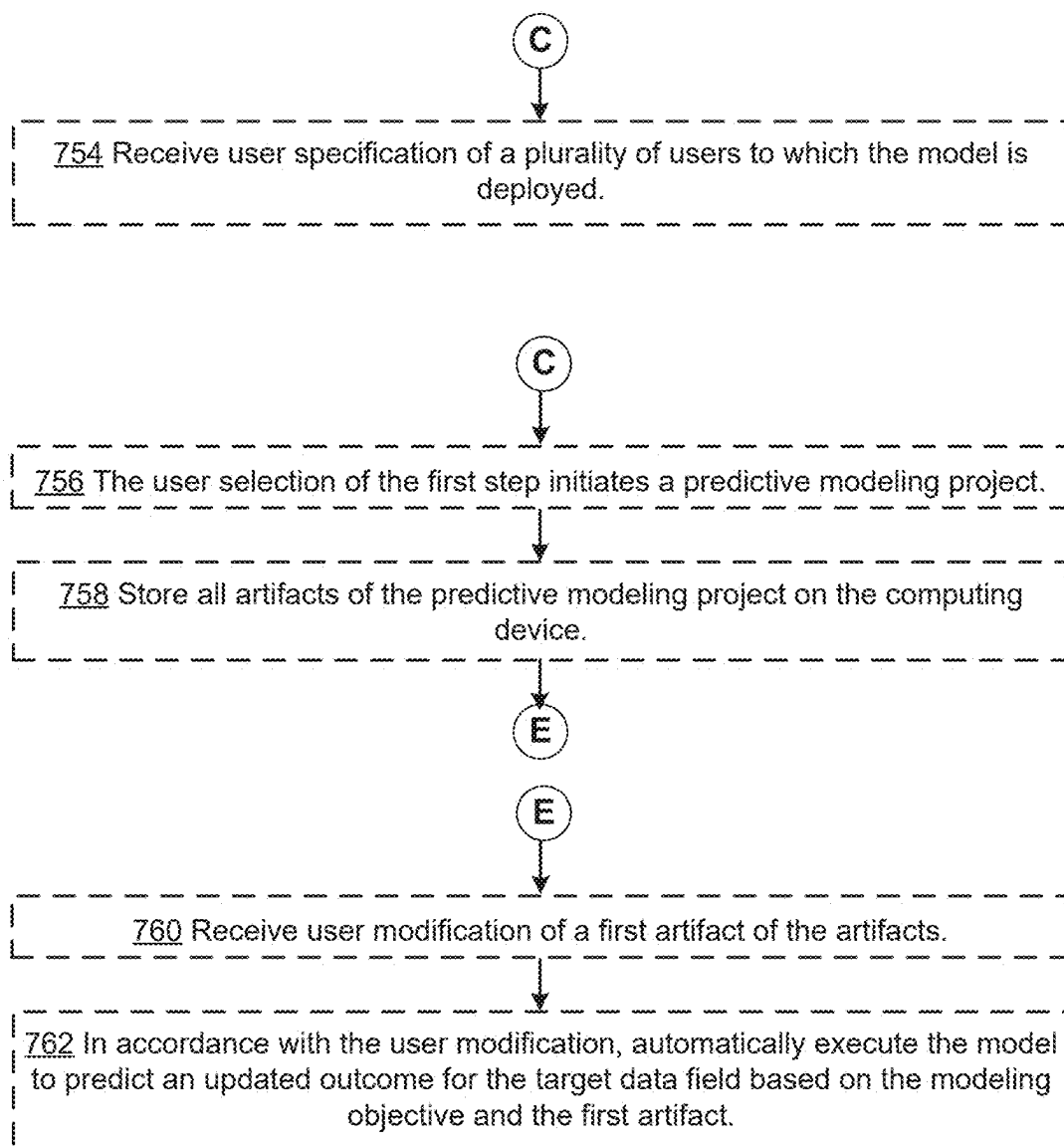


Figure 7D

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**USER INTERFACE FOR MACHINE
LEARNING MODEL GENERATION****RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Patent Application No. 63/242,012, filed Sep. 8, 2021, entitled “Visual Tracing and Editing of Machine Learning Models,” which is hereby incorporated by reference herein in its entirety.

This application is related to “Introduction to Einstein Discovery,” available at https://help.salesforce.com/s/articleView?id=sf.bi_edd_about.htm&type=5, which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

The disclosed implementations relate generally to data visualization and more specifically to systems, methods, and user interfaces for interactive visual analysis of a data set.

BACKGROUND

Data visualization applications enable a user to understand a data set visually. Visual analyses of data sets, including distribution, trends, outliers, and other factors are important to making business decisions. Some data sets are very large or complex, and include many data fields. Some data elements are computed based on data from a selected data set. Various tools can be used to help understand and analyze the data, including dashboards that have multiple data visualizations and natural language interfaces that help with visual analytical tasks.

Today, adding predictions from machine learning (ML) models to a data visualization application requires users to build custom ML models outside of the data visualization application using custom ML applications or other tools. This creates a disjointed, multi-product experience and requires users to learn and leverage different product user interfaces. This disjointed experience also siloes predictive model builders (e.g., data analysts) away from the teams that need predictions to make decisions. The segregation between data analysts and the teams leads to slow delivery of results, introduces errors due to misunderstood requirements, and can render even successful predictive models irrelevant as the world changes.

SUMMARY

There is a need for improved systems and methods that support interactions with data visualization (e.g., visual analytics) systems. The present disclosure describes systems, user interfaces, methods, and devices that integrate (e.g., combine) visual analytics with predictive analytics. For example, data that was previously generated using a data visualization application can be augmented with statistical modeling and machine learning (ML), and turned into predictive ML models that can identify, surface, and visualize insights into the data. The models and predictions can be presented in a manner that is easily understood and visualized by users. Users can evaluate the performance of a trained ML model, understand what variables have the greatest impact on a modeling objective, and determine how robust the model is likely to be on future data it has yet to see. If a model meets the user’s expectations, the user can deploy the model and make the model available to other users across their organization to generate predictions. Once

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deployed, anyone with access to the modeling project can track the model’s performance and utilization over time.

The present disclosure describes systems and user interfaces that enable data scientists/analysts (also known as “Business Scientists”) to collaborate with other members within an organization (who may not have a data science background) to train, understand, evaluate, and operationalize no-code predictive ML models. For example, instead of extracting data away from the business domain to start modeling, a business scientist can create a modeling project that includes all relevant stakeholders. Together, these team members can clarify business questions and identify relevant data that, with a series of click-through steps, can fit an ML model to generate predictions to improve decisions. Collaborators can use visualizations to understand the effectiveness of the model, review potential bias problems, and explore how features drive predictions. Thus, by democratizing access to predictive analytics, organizations are empowered to accelerate and improve decisions. Today, these are missed opportunities due to the limited bandwidth of centralized data science teams. Augmenting existing analyses with predictions built close to the business delivers faster, more complete analytics supporting quicker, more accurate decisions.

In some implementations of the present disclosure, once a trained model is deployed, the modeling project exists in a data catalog, thereby providing an auditable trail for all data and modeling choices. Business scientists can monitor and improve the model’s performance. They can also identify new problems to be solved. The disclosed ML models can integrate predictions into data visualization dashboards, worksheets, and data prep workflows in accordance with some implementations.

In accordance with some implementations, a method generates predictive analytics based on no-code machine learning (ML) models. The method is performed at a computing device that includes a display, one or more processors, and memory. The memory stores one or more programs configured for execution by the one or more processors. The method includes displaying, in a user interface, a workflow that includes a plurality of steps. The method is responsive to user selection of a first step of the plurality of steps, displaying a list of data sources. The method receives user selection of a first data source of the data sources. The method then receives user input specifying a target data field from the first data source and a modeling objective for the target data field. In response to the user input, the method automatically executes a model to predict a first outcome for the target data field based on the modeling objective and displays results of the model. The method also receives user input to deploy the model. The method then deploys the model.

In some implementations, each of the steps of the workflow is a user-selectable element in the user interface.

In some implementations, the first step corresponds to a first user-selectable element. The method further comprises, in response to user selection of the first element, displaying the first element in a visually distinct manner from other elements corresponding to other steps in the workflow.

In some implementations, after receiving the user input specifying the target data field and the modeling objective, the method updates the first element to indicate completion of the first step.

In some implementations, in response to the user input specifying the target data field, the method determines one or more second data fields from the data source and a respective correlation between the target data field and each

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of the one or more second data fields. The method displays the respective correlations as a ranked bar chart in the user interface.

In some implementations, the results of the model are displayed on a side pane of the user interface.

In some implementations, displaying the results of the model includes displaying one or more of: a model title, a date of generation of the model, a date of update of the model, and a version of the model.

In some implementations, displaying the results of the model includes displaying a plurality of metrics of the model.

In some implementations, the method further comprises receiving user selection of a first metric of the plurality of metrics. In response to the user selection, the method displays a plurality of navigation tabs in the user interface. User selection of a respective tab causes respective information about the first metric to be displayed.

In some implementations, the method further comprises displaying a progress bar for visualizing progress of the model execution.

In some implementations, the method further comprises receiving user specification of a plurality of users to which the model is to be deployed.

In some implementations, the user selection of the first step initiates a predictive modeling project. The method further comprises storing all artifacts of the predictive modeling project on the computing device.

In some implementations, the method further comprises receiving user modification of a first artifact of the artifacts. The method also comprises, in accordance with the user modification, automatically executing the model to predict an updated outcome for the target data field based on the modeling objective and the first artifact.

In some implementations, deploying the model comprises sending the model to a plurality of users via a plurality of modes of communication.

In some implementations, a computing device includes a display, one or more processors, memory, and one or more programs stored in the memory. The programs are configured for execution by the one or more processors. The one or more programs include instructions for performing any of the methods described herein.

In some implementations, a non-transitory computer-readable storage medium stores one or more programs configured for execution by a computing device having one or more processors and memory. The one or more programs include instructions for performing any of the methods described herein.

Thus methods, systems, and graphical user interfaces are disclosed that enable users to easily integrate predictions from machine learning (ML) models with data visualizations.

BRIEF DESCRIPTION OF THE DRAWINGS

For a better understanding of the aforementioned systems, methods, and graphical user interfaces, as well as additional systems, methods, and graphical user interfaces that provide data visualization analytics, reference should be made to the Description of Implementations below, in conjunction with the following drawings in which like reference numerals refer to corresponding parts throughout the figures.

FIG. 1 shows a graphical user interface used in some implementations.

FIG. 2 is a block diagram of a computing device according to some implementations.

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FIG. 3 is a block diagram of a server system according to some implementations.

FIG. 4 illustrates an example server architecture 400 for predictive analytics according to some implementations.

FIGS. 5A and 5B illustrate a user interface 500 for data preparation according to some implementations.

FIGS. 6A-6AA illustrate a series of screenshots of a graphical user interface 600 for a machine learning (ML) workflow, in accordance with some implementations.

FIGS. 7A-7D provide a flowchart of a method performed at a computing device according to some implementations.

Reference will now be made to implementations, examples of which are illustrated in the accompanying drawings. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. However, it will be apparent to one of ordinary skill in the art that the present invention may be practiced without requiring these specific details.

DESCRIPTION OF IMPLEMENTATIONS

The disclosed implementations integrate (e.g., combine) a ML platform with a visual analytics platform (e.g., data visualization application) to create a new collaborative experience around how to discover, transform, visualize, iterate, and publish new models on their data.

The disclosed methods, systems, and user interfaces allow teams to collaboratively translate data (e.g., data sources that reside on the visual analytics platform) and their domain knowledge into better decisions augmented by leading-edge ML models.

In some implementations, business scientists initiate a collaborative experience by creating a modeling project built around a simple flow-based user interface (UI) and inviting team members to discuss the problem and share input throughout the process. The team can choose existing data sources from the visual analytics platform (e.g., Tableau) or use data preparation web authoring to prepare training data for the project. Users (e.g., team members) can then discuss and describe their business question by selecting a data field and an objective (e.g., whether they want to increase it (e.g., sales, profit, conversions) or decrease it (e.g., churn, defects)). In some implementations, data is automatically transmitted to the ML platform, to fit an optimal model to the user's business requirements.

In some implementations, the ML platform augments the modeling process by visually surfacing statistically significant patterns, which can be shared as Tableau workbooks. Teams use these insights to explore the model and rapidly iterate to improve accuracy. Throughout the process, a user experience centered on model interpretability and automated bias detection helps collaborators understand model results and potential pitfalls in plain business language.

In some implementations, after determining a model is safe and useful, users can publish it to start generating predictions. Predictions can be consumed in data and visualizations, tracked for performance and data drift, and integrated into the data catalog all inside Tableau.

In some implementations, the users of an integrated predictive and visual analytics platform include:

- Data analysts/advanced business users who can navigate through data insights and model metrics;

- Business users who will be consuming the outputs of the predictive models and validating that the results of the predictions are providing the level of value for their business that they expected;

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Operations teams who understand how the business functions;

IT teams who might need to help wrangle additional data or be required to maintain the models; and/or

Data scientists who can collaborate and provide guidance on modeling projects to ensure things like auditing, assumptions, and modeling decisions are accurate

FIG. 1 shows a graphical user interface **100** for interactive data analysis according to some implementations. The user interface **100** includes a Data tab **114** and an Analytics tab **116** in accordance with some implementations. When the Data tab **114** is selected, the user interface **100** displays a schema information region **110**, which is also referred to as a data pane. The schema information region **110** provides named data elements (e.g., field names) that may be selected and used to build a data visualization. In some implementations, the list of field names is separated into a group of dimensions (e.g., categorical data) and a group of measures (e.g., numeric quantities). Some implementations also include a list of parameters. When the Analytics tab **116** is selected, the user interface displays a list of analytic functions instead of data elements (not shown).

The graphical user interface **100** also includes a data visualization region **112**. The data visualization region **112** includes a plurality of shelf regions, such as a columns shelf region **120** and a rows shelf region **122**. These are also referred to as the column shelf **120** and the row shelf **122**. As illustrated here, the data visualization region **112** also has a large space for displaying a visual graphic (also referred to herein as a data visualization). Because no data elements have been selected yet, the space initially has no visual graphic. In some implementations, the data visualization region **112** has multiple layers that are referred to as sheets. In some implementations, the data visualization region **112** includes a region **126** for data visualization filters.

In some implementations, the shelf regions determine characteristics of a desired data visualization. For example, a user can place field names into these shelf regions (e.g., by dragging fields from the schema information region **110** to the column shelf **120** and/or the row shelf **122**), and the field names define the data visualization characteristics. A user may choose a vertical bar chart, with a column for each distinct value of a field placed in the column shelf region. The height of each bar is defined by another field placed into the row shelf region.

In some implementations, the graphical user interface **100** includes a natural language input box **124** (also referred to as a command box) for receiving natural language commands. A user may interact with the command box to provide commands. For example, the user may provide a natural language command by typing in the box **124**. In addition, the user may indirectly interact with the command box by speaking into a microphone **220** to provide commands. In some implementations, data elements are initially associated with the column shelf **120** and the row shelf **122** (e.g., using drag and drop operations from the schema information region **110** to the column shelf **120** and/or the row shelf **122**). After the initial association, the user may use natural language commands (e.g., in the natural language input box **124**) to further explore the displayed data visualization. In some instances, a user creates the initial association using the natural language input box **124**, which results in one or more data elements being placed on the column shelf **120** and on the row shelf **122**. For example, the user may provide a command to create a relationship between a data element X and a data element Y. In response to receiving the command, the column shelf **120** and the row

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shelf **122** may be populated with the data elements (e.g., the column shelf **120** may be populated with the data element X and the row shelf **122** may be populated with the data element Y, or vice versa).

In some implementations, the graphical user interface **100** includes a view level detail icon **128**, which can be used to specify or modify the level of detail for the data visualization. The view level detail icon **128** enables a user to specify a level of detail that applies to the data visualization overall or to specify additional fields that will be included in the overall level of detail (in addition to those that are included by default). Typically, implementations have only one “overall” level of detail. Other levels of detail may be specified within individual contexts, as described below.

In some implementations, the graphical user interface **100** includes an encodings region **130** to specify various encodings for a data visualization.

FIG. 2 is a block diagram illustrating a computing device **200** that can display the graphical user interface **100** or the predictive analytics UI **600** in accordance with some implementations. The computing device can also be used by a data preparation (“data prep”) application **230** or a predictive analytics application **250**. Various examples of the computing device **200** include a desktop computer, a laptop computer, a tablet computer, and other computing devices that have a display and a processor capable of running a data visualization application **222**, a data prep application **230**, and/or a predictive analytics application **250**. The computing device **200** typically includes one or more processing units/cores (CPUs) **202** for executing modules, programs, and/or instructions stored in the memory **214** and thereby performing processing operations, one or more network or other communications interfaces **204**, memory **214**, and one or more communication buses **212** for interconnecting these components. The communication buses **212** may include circuitry that interconnects and controls communications between system components.

The computing device **200** includes a user interface **206** comprising a display device **208** and one or more input devices or mechanisms **210**. In some implementations, the input device/mechanism includes a keyboard. In some implementations, the input device/mechanism includes a “soft” keyboard, which is displayed as needed on the display device **208**, enabling a user to “press keys” that appear on the display **208**. In some implementations, the display **208** and input device/mechanism **210** comprise a touch screen display (also called a touch sensitive display).

In some implementations, the memory **214** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM or other random access solid state memory devices. In some implementations, the memory **214** includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. In some implementations, the memory **214** includes one or more storage devices remotely located from the CPU(s) **202**. The memory **214**, or alternatively the non-volatile memory devices within the memory **214**, comprises a non-transitory computer readable storage medium. In some implementations, the memory **214**, or the computer readable storage medium of the memory **214**, stores the following programs, modules, and data structures, or a subset thereof:

- an operating system **216**, which includes procedures for handling various basic system services and for performing hardware dependent tasks;

- a communications module **218**, which is used for connecting the computing device **200** to other computers

and devices via the one or more communication network interfaces **204** (wired or wireless) and one or more communication networks, such as the Internet, other wide area networks, local area networks, metropolitan area networks, and so on;

a web browser **220** (or other application capable of displaying web pages), which enables a user to communicate over a network with remote computers or devices;

a data visualization application **222**, which provides a graphical user interface **100** for a user to construct visual graphics. For example, a user selects one or more data sources **240** (which may be stored on the computing device **200** or stored remotely, such as on a server system **300**), selects data fields from the data sources, and uses the selected fields to define a visual graphic. In some implementations, the information the user provides is stored as a visual specification **228**. The data visualization application **222** includes a data visualization generation module **226**, which takes the user input (e.g., the visual specification **228**), and generates a corresponding visual graphic (also referred to as a “data visualization” or a “data viz”). The data visualization application **222** then displays the generated visual graphic in the user interface **100**. In some implementations, the data visualization application **222** executes as a standalone application (e.g., a desktop application). In some implementations, the data visualization application **222** executes within the web browser **220** or another application using web pages provided by a web server;

zero or more databases or data sources **240** (e.g., a first data source **240-1**), which are used by the data visualization application **222**. In some implementations, the data sources are stored as spreadsheet files, CSV files, XML files, or flat files, or stored in a relational database;

zero or more semantic models **242** (e.g., a first semantic model **242-1**), each of which is derived directly from a respective database or data source **240**. A semantic model **242** represents the database schema and contains metadata about attributes. In some implementations, a semantic model **242** also includes metadata of alternative labels or synonyms of the attributes. A semantic model **242** includes data types (e.g., “text,” “date,” “geospatial,” “Boolean,” and “numeric”), attributes, (e.g., a currency type such as the United States Dollar), and a semantic role or a data role (e.g., “City” role for a geospatial attribute) for data fields of the respective database or data source **240**. In some implementations, a semantic model **242** also captures statistical values (e.g., data distribution, range limits, average, and cardinality) for each attribute. In some implementations, a semantic model **242** is augmented with a grammar lexicon that contains a set of analytical concepts found in many query languages (e.g., average, filter, and sort). In some implementations, a semantic model **242** also distinguishes between attributes that are measures (e.g., attributes that can be measured, aggregated, or used for mathematical operations) and dimensions (e.g., fields that cannot be aggregated except by counting). In some implementations, the semantic models **242** include one or more concept graphs that encapsulate semantic information for the data sources **240**. In some implementations, the one or more concept graphs are organized as directed acyclic graphs, and/or embody hierarchical inheritance of semantics between one or more

entities (e.g., logical fields, logical tables, and data fields). Thus, the semantic models **242** help with inferring and assigning semantic roles to fields;

one or more object models **108**, which identify the structure of the data sources **240**. In an object model (or a data model), the data fields (attributes) are organized into classes, where the attributes in each class have a one-to-one correspondence with each other. The object model also includes many-to-one relationships between the classes. In some instances, an object model maps each table within a database to a class, with many-to-one relationships between classes corresponding to foreign key relationships between the tables. In some instances, the data model of an underlying source does not cleanly map to an object model in this simple way, so the object model includes information that specifies how to transform the raw data into appropriate class objects. In some instances, the raw data source is a simple file (e.g., a spreadsheet), which is transformed into multiple classes;

In some implementations, the computing device **200** stores a data prep application **230**, which can be used to analyze and massage data for subsequent analysis (e.g., by a data visualization application **222**). FIG. 5 illustrates one example of a data prep user interface **500**. The data prep application **230** enables user to build flows **523**, as described in more detail below.

In some implementations, the computing device **200** stores a predictive analytics application **250**. The predictive analytics application **250** includes a predictive analytics generation module **252**, which takes the user input (e.g., user selection of a data source **240**, a data field of the data source, and a modeling objective) and automatically executes a no-code machine learning (ML) model to build a workflow that delivers predictive analytics. The predictive analytics application **250** provides a predictive analytics user interface **600** for a user to select one or more predictive models **254** (e.g., machine learning models) (e.g., a first predictive model **254-1**) and generate predictions using the prediction model, based on input from the user and historical data of a data source **240**. In some implementations, the predictive analytics application **250** integrates predictions from the models into table calculations, dashboard extensions, and Prep flows, as described in more detail below. In some implementations, the ML model building workflow enables business teams to do “data science as a team sport” and collaborate to deliver predictive analytics that are informed by their business domain knowledge and can be easily integrated into existing business processes.

In some implementations, the computing device **200** stores data **256** that is generated during the predictive modeling, such as model results **258** and/or artifacts **260**. As used herein, an artifact **260** is an item generated and exchanged by human or machine actions across an end-to-end automated data science (ML) workflow, which comprises preparation, analysis, deployment, and communication stages. For example, the predictive model **254**, descriptive statistics about the model performance, correlations between a target data field and other data fields of a data source **240**, data changes that are made (e.g., removal of one or more data columns from the data source or filtering data rows) are artifacts generated in a predictive modeling project.

In some implementations, the computing device **200** stores APIs **262** for receiving API calls from one or more applications (e.g., from a web browser **220**, a data visualization application **222**, a data prep application **230**, a

predictive analytics application **250**, a data visualization web application **322**, or a predictive analytics web application **334**), translating the API calls into appropriate actions, and performing one or more actions.

In some implementations, the computing device **200** includes a widget generation module, which generates widgets that include user-selectable options. For example, a “sort” widget is generated in response to a user selecting (e.g., hovering) over a sort field (e.g., a natural language term identified to be a sort field). The sort widget includes user-selectable options such as “ascending,” “descending,” and/or “alphabetical,” so that the user can easily select, from the widget, how to sort the selected field.

Each of the above identified executable modules, applications, or sets of procedures may be stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various implementations. In some implementations, the memory **214** stores a subset of the modules and data structures identified above. Furthermore, the memory **214** may store additional modules or data structures not described above.

Although FIG. **2** shows a computing device **200**, FIG. **2** is intended more as a functional description of the various features that may be present rather than as a structural schematic of the implementations described herein. In practice, and as recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated.

FIG. **3** is a block diagram illustrating an example server system **300** in accordance with some implementations. In some implementations, the server system **300** is a data visualization server and/or a predictive analytics server. A server system **300** typically includes one or more processing units/cores (CPUs) **302**, one or more network interfaces **304**, memory **314**, and one or more communication buses **312** for interconnecting these components. In some implementations, the server system **300** includes a user interface **306**, which includes a display **308** and one or more input devices **310**, such as a keyboard and a mouse. In some implementations, the communication buses **312** includes circuitry (sometimes called a chipset) that interconnects and controls communications between system components.

In some implementations, the memory **314** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. In some implementations, the memory **314** includes one or more storage devices remotely located from the CPUs **302**. The memory **314**, or alternatively the non-volatile memory devices within the memory **314**, comprise a non-transitory computer readable storage medium.

In some implementations, the memory **314** or the computer readable storage medium of the memory **314** stores the following programs, modules, and data structures, or a subset thereof:

- an operating system **316**, which includes procedures for handling various basic system services and for performing hardware dependent tasks;
- a network communication module **318**, which is used for connecting the server **300** to other computers via the

one or more communication network interfaces **304** (wired or wireless) and one or more communication networks, such as the Internet, other wide area networks, local area networks, metropolitan area networks, and so on;

- a web server **320** (such as an HTTP server), which receives web requests from users and responds by providing responsive web pages or other resources;
- a data visualization web application **322**, which may be downloaded and executed by a web browser **226** on a user’s computing device **200**. In general, a data visualization web application **322** has the same functionality as a desktop data visualization application **222**, but provides the flexibility of access from any device at any location with network connectivity, and does not require installation and maintenance. In some implementations, the data visualization web application **322** includes various software modules to perform certain tasks. In some implementations, the web application **322** includes a user interface module **324**, which provides the user interface for all aspects of the web application **322**. In some implementations, the web application **322** includes a data retrieval module **326**, which builds and executes queries to retrieve data from one or more data sources **242**. The data sources **240** may be stored locally on the server system **300** or stored in an external database **328**. In some implementations, data from two or more data sources may be blended. In some implementations, the data retrieval module **326** uses a visual specification **228** to build the queries. In some implementations, the visual specification includes one or more aggregate specifications;
- a predictive analytics web application **334**, which may be downloaded and executed by a web browser **226** on a user’s computing device **200**. In general, a predictive analytics web application **334** has the same functionality as a desktop predictive analytics application **250**, but provides the flexibility of access from any device at any location with network connectivity, and does not require installation and maintenance. In some implementations, the predictive analytics web application **334** includes various software modules to perform certain tasks. In some implementations, the predictive analytics web application **334** includes a user interface module **335**, which provides the user interface for all aspects of the predictive analytics web application **334**. In some implementations, the predictive analytics web application **334** includes a predictive analytics generation module **336**, which takes user input (e.g., user selection of a data source **240**, a data field **330** of the data source **240**, a modeling objective) and automatically executes a no-code machine learning (ML) model building workflow to deliver predictive analytics. In some implementations, the predictive analytics web application **334** includes predictive models **254** (e.g., machine learning models) such as predictive model **1 254-1**, which is trained to generate predictions based on user input and historical data of a data source **240**;
- one or more databases **328**, which store data used or created by the data visualization web application **322**, the data visualization application **230**, the predictive analytics web application **334**, the predictive analytics application **250**, or the data prep application **230**. The databases **328** may store data sources **240**, which provide the data used in the generated data visualizations or predictive analytics. Each data source **240** includes one or more data fields **330**. In some imple-

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mentations, the database 328 stores user preferences 332. In some implementations, the database 328 includes a data visualization history log, which tracks each time the data visualization renders a data visualization. In some implementations, the database 328 stores predictive model results 258. In some implementations, the database 328 stores predictive model results 258 by modeling projects. In some implementations, the database 328 stores artifacts 260;

object models 108; and

APIs 338 for receiving API calls from one or more applications (e.g., the web browser 220, the data visualization application 222, the data prep application 230, the predictive analytics application 250, the data visualization web application 322, or the predictive analytics web application 334), translating the API calls into appropriate actions, and performing one or more actions.

The databases 328 may store data in many different formats, and commonly includes many distinct tables, each with a plurality of data fields 330. Some data sources comprise a single table. The data fields 330 include both raw fields from the data source (e.g., a column from a database table or a column from a spreadsheet) as well as derived data fields, which may be computed or constructed from one or more other fields. For example, derived data fields include computing a month or quarter from a date field, computing a span of time between two date fields, computing cumulative totals for a quantitative field, computing percent growth, and so on. In some instances, derived data fields are accessed by stored procedures or views in the database. In some implementations, the definitions of derived data fields 330 are stored separately from the data source 240. In some implementations, the database 328 stores a set of user preferences 332 for each user. The user preferences may be used when the data visualization web application 322 (or application 222) makes recommendations about how to view a set of data fields 330. In some implementations, the database 328 stores a data visualization history log, which stores information about each data visualization generated. In some implementations, the database 328 stores other information, including other information used by the data visualization application 222, the data visualization web application 322, the predictive analytics application 250, or the predictive analytics web application 334. The databases 328 may be separate from the server system 300, or may be included with the server system (or both). In some implementations, the database stores semantic models as described above for the computing device 200.

In some implementations, the database 328 includes a data visualization history log and/or a visual analytics history log, which stores visual specifications 228 and/or predictive models selected by users. The history log may include a user identifier, a timestamp of when the data visualization and/or predictive model was created, a list of the data fields used in the data visualization and/or predictive model, the type of the data visualization (sometimes referred to as a “view type” or a “chart type”), data encodings (e.g., color and size of marks), the data relationships selected, and what connectors are used. In some implementations, one or more thumbnail images of each data visualization are also stored. Some implementations store additional information about created data visualizations, such as the name and location of the data source, the number of rows from the data source that were included in the data visualization, the version of the data visualization software, and so

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on. Some implementations store a version number and/or version history of trained (e.g., generated) predictive models.

Each of the above identified executable modules, applications, or sets of procedures may be stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various implementations. In some implementations, the memory 314 stores a subset of the modules and data structures identified above. Furthermore, the memory 314 may store additional modules or data structures not described above.

Although FIG. 3 shows a server system 300, FIG. 3 is intended more as a functional description of the various features that may be present rather than as a structural schematic of the implementations described herein. In practice, and as recognized by those of ordinary skill in the art, items shown separately could be combined and some items could be separated. In addition, some of the programs, functions, procedures, or data shown above with respect to a server system 300 may be stored or executed on a computing device 200. In some implementations, the functionality and/or data may be allocated between a computing device 200 and one or more servers 300. Furthermore, one of skill in the art recognizes that FIG. 3 need not represent a single physical device. In some implementations, the server functionality is allocated across multiple physical devices that comprise a server system. As used herein, references to a “server” include various groups, collections, or arrays of servers that provide the described functionality, and the physical servers need not be physically colocated (e.g., the individual physical devices could be spread throughout the United States or throughout the world).

FIG. 4 illustrates an example server architecture 400 for predictive analytics according to some implementations. In some implementations, the server architecture 400 includes a first server 410 (e.g., the Salesforce L&P server), a second server 420 (e.g., a Tableau CRM server), and a third server 430 (e.g., a Tableau server) that are communicatively connected with one another. In some implementations, the first server 410 and the second server 420 include APIs 338-1, 338-2, and 338-3 for receiving API calls from one or more applications and/or services residing on the third server 430.

In some implementations, the first server 410 (e.g., the Salesforce L&P server) stores metadata 411 that describes licensing and permissions data.

In some implementations, the second server 420 (e.g., the Tableau CRM server) includes an Einstein Discovery module 414 for making predictions and recommendations based on datasets and their historical trends. The Einstein Discovery module 414 can determine how past data events relate to a key performance indicator (KPI) in question (also known as an “Outcome”). In some implementations, the Einstein Discovery module 414 uses models 254 (e.g., predictive models, machine learning models, or sophisticated custom equations) to predict a particular outcome. Each of the models 254 accepts inputs (one or more explanatory variables) and produces outputs (e.g., a predicted outcome, top factors, and improvements). In some implementations, relationships between past data events and outcomes are depicted as stories 416 and stored on the second server 420. In some implementations, the Einstein Discovery module 414 trains the models 254 using data (e.g., from datasets

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and/or data sources **240**), which may be stored either on the second server **420** or on other servers, such as the third server **430** (e.g., Tableau server) or a data visualization server. In some implementations, the model training datasets **418** are stored on the second server **420**.

In some implementations, the third server **430** (e.g., the Tableau server) includes a Prometheus microservice **402** for providing services for predictive analytics and visual analysis. In some implementations, the Prometheus microservice **402** provides an authoring UI **404** for authoring predictive analytics modeling projects. In accordance with some implementations, the Prometheus Microservice **402** receives a request from a user for initiating a predictive modeling project. In response to the request, the third server **430** (e.g., Prometheus microservice **402**) can send an API call to the first server **410** (e.g., via the first API **338-1**), to have the first server **410** verify (e.g., using the metadata **411**) whether the user has the required license and/or credentials for initiating the project. In some implementations, the Prometheus microservice **402** includes a model management UI **406**, which can be used to update model metadata of list models coming from Einstein Discovery.

In some implementations, as illustrated in FIG. 4, the authoring UI **404** includes other interfaces, such as a Prep UI **422** (which originated from a data prep module **412**) and/or a visualization client UI **424** (which originated from the monolith module **426**).

In some implementations, the Prometheus microservice **402** interacts with the Einstein Discovery module **414** via the first API **338-1** to obtain predictive modeling insights, update model metadata (e.g., artifacts), start model training, and/or monitor model training.

In some implementations, the third server **430** includes a scoring integrations module **408** for performing scoring operations. The scoring operations can include consolidating data in a dataset and/or data source, creating calculations based on data values of data fields in a data source **240**, and/or preparing data such as removing extremes, outliers, missing values, incorrect values, skew and cardinality. In some implementations, the scoring operations are performed before model training so as to optimize the data for machine learning and modeling.

User Interface for Data Prep

FIGS. 5A and 5B illustrate a user interface **500** for data preparation according to some implementations.

FIG. 5A illustrates an overview of the user interface **500** for data preparation, showing panes that group together different functionality. In some implementations, the left-hand pane **512** provides options for a user to locate and connect to data or to perform operations on already selected data. In some implementations the flow area **513** illustrates one or more operations at nodes to perform on selected data (e.g., data manipulations to prepare data for analysis). In some implementations the profile area **514** provides information about the data set at the currently selected node (e.g., histograms of data value distributions for some of the data fields in the data set). In some implementations, the data grid **515** provides raw data values in rows and columns of the data set at the currently selected node.

FIG. 5B provides a concrete example of a user interface **500** for data preparation, showing the user interface elements in each of the panes. The menu bar **511** includes one or more menus, such as a File menu and an Edit menu. Although the edit menu is available, more changes to the flow are performed by interacting with the flow pane **513**, the profile pane **514**, or the data pane **515**.

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In some implementations, the left-hand pane **512** includes a data source palette/selector. The left-hand pane **512** also includes an operations palette, which displays operations that can be placed into the flow. In some implementations, the list of operations includes arbitrary joins (of arbitrary type and with various predicates), union, pivot, rename and restrict column, projection of scalar calculations, filter, aggregation, data type conversion, data parse, coalesce, merge, split, aggregation, value replacement, and sampling. Some implementations also support operators to create sets (e.g., partition the data values for a data field into sets), binning (e.g., grouping numeric data values for a data field into a set of ranges), and table calculations (e.g., calculate data values, such as percent of total, for each row, which depends not only on the data values in each row, but also on other data values in the table).

The left-hand pane **512** also includes a palette of other flows that can be incorporated in whole or in part into the current flow. This enables a user to reuse components of a flow to create new flows. For example, if a portion of a flow has been created that scrubs a certain type of input using a combination of 10 steps, that 10 step flow portion can be saved and reused, either in the same flow or in completely separate flows.

The flow pane **513** displays a visual representation (e.g., a node/link flow diagram) **523** for the current flow. The Flow Pane **513** provides an overview of the flow, which serves to document the process. As the number of nodes increases, implementations typically add scroll boxes. The need for scroll bars is reduced by coalescing multiple related nodes into super nodes, which are also called container nodes. This enables a user to see the entire flow more conceptually, and allows a user to dig into the details only when necessary. In some implementations, when a “super node” is expanded, the flow pane **513** shows just the nodes within the super node, and the flow pane **513** has a heading that identifies what portion of the flow is being displayed. Implementations typically enable multiple hierarchical levels.

A complex flow is likely to include several levels of node nesting. Different nodes within the flow diagram **523** perform different tasks, and thus the node internal information is different. In addition, some implementations display different information depending on whether or not a node is selected. A flow diagram **523** provides an easy, visual way to understand how the data is getting processed, and keeps the process organized in a way that is logical to a user.

In some implementations, the profile pane **514** includes schema information about the data set at the currently selected node (or nodes) in the flow pane **513**. As illustrated here, the schema information provides statistical information about the data, such as a histogram **524** of the data distribution for each of the fields. A user can interact directly with the profile pane to modify the flow **523** (e.g., by selecting a data field for filtering the rows of data based on values of that data field). The profile pane **514** also provides users with relevant data about the currently selected node (or nodes) and visualizations that guide a user’s work. For example, the histograms **524** show the distributions of the domains of each column. Some implementations use brushing to show how these domains interact with each other.

The data pane **515** displays the rows **525** of data corresponding to the selected node or nodes in the flow pane **513**. Each of the columns **526** corresponds to one of the data fields. A user can interact directly with the data in the data pane to modify the flow **523** in the flow pane **513**. A user can also interact directly with the data pane to modify individual field values. In some implementations, when a user makes a

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change to one field value, the user interface applies the same change to all other values in the same column whose values (or pattern) match the value that the user just changed.

The sampling of data in the data pane **515** is selected to provide valuable information to the user. For example, some implementations select rows that display the full range of values for a data field (including outliers). As another example, when a user has selected nodes that have two or more tables of data, some implementations select rows to assist in joining the two tables. The rows displayed in the data pane **515** are selected to display both rows that match between the two tables as well as rows that do not match. This can be helpful in determining which fields to use for joining and/or to determine what type of join to use (e.g., inner, left outer, right outer, or full outer).

Although a user can edit a flow diagram **523** directly in the flow pane **513**, changes to the operations are typically done in a more immediate fashion, operating directly on the data or schema in the profile pane **514** or the data pane **515** (e.g., right clicking on the statistics for a data field in the profile pane to add or remove a column from the flow).

User Interface for ML Workflow

In accordance with some implementations of the present disclosure, an experience that allows teams to collaboratively translate the historical data they have connected to a data analytics application (e.g., Tableau), and their domain knowledge, into better decisions augmented by leading-edge ML models can include the following steps:

Business scientists start by creating a modeling project built around a simple flow-based UI and inviting team members to discuss the problem and share input throughout the process;

The team chooses an existing data source **240** (e.g., on a computing device or on a server system such as a data visualization server, the server **300**, the second server **420**, or the third server **430**) or uses Prep web authoring to prepare training data for the project;

Users then discuss and describe their business question by selecting a field in the data and a modeling objective corresponding to the field (e.g., whether they want to increase it (e.g., sales, profit, or conversions) or decrease it (e.g., churn or defects)). In some implementations, the data automatically flows to a server (e.g., the second server **420**, in the Einstein Discovery module **414**) to fit an optimal model to the user's business requirements;

In some implementations, the second server **420** (e.g., the Einstein Discovery module **414**) augments the modeling process by visually surfacing statistically significant patterns, which can be shared as Tableau workbooks. Teams use these insights to explore the model and rapidly iterate to improve accuracy;

Throughout the process, a user experience centered on model interpretability and automated bias detection helps collaborators understand model results and potential pitfalls in plain business language; and

In some implementations, after determining that a model is safe and useful, users can publish it to start generating predictions. Predictions can be consumed in data and data visualizations, tracked for performance and data drift, and integrated into a data catalog that is stored on a server system (e.g., a data visualization server, the server **300**, the second server **420**, or the third server **430**).

FIGS. 6A-6AA illustrate a series of screenshots of a graphical user interface **600** for a machine learning (ML) workflow, in accordance with some implementations. In

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some implementations, the graphical user interface **600** corresponds to the authoring UI **404**, which executes on a computing device **200**. In some implementations, the graphical user interface **600** is an interface that supports integrated visual and predictive analytics.

In some implementations, the experience starts with a user (e.g., a project creator, such as a business scientist) creating a modeling project. FIG. 6A illustrates user selection of an icon **602** (e.g., a "New" icon). In response to user selection of the icon **602**, the graphical user interface **600** displays a dropdown menu **604**. The user can select a "Prediction Model" **606** option to create (e.g., initiate) a new modeling project.

In some implementations, in response to the creation of a modeling project, the graphical user interface **600** displays a workflow **610** (e.g., a ML workflow, a workflow interface, or a workflow pipeline) as illustrated in FIG. 6B. The workflow **610** includes blocks **612**, each corresponding to a step of the workflow. For example, the steps can include (i) "Connect to Data" (e.g., connect to a data source), represented by the first block **612-1**, (ii) "Define objective" (e.g., defining a modeling objective), represented by the second block **612-2**, (iii) "Train Model," represented by the fourth block **612-3**, (iv) "Model Results," represented by the fourth block **612-4**, (v) "Deploy Model," represented by the fifth block **612-5**, and (vi) "Track Deployment," represented by the sixth block **612-6**.

In some implementations, as illustrated in FIG. 6B, the workflow **610** includes lines **614** connecting adjacent blocks **612**.

In some implementations, each of the blocks **612** (e.g., steps) of the workflow **610** is a user-selectable element in the graphical user interface **600**. In some implementations, at the start of the modeling project, the graphical user interface **600** displays each of the blocks **612** as grayed out (e.g., as an inactive or visually de-emphasized) and each of the lines **614-1**, **614-2**, **614-3**, **614-4**, and **614-5** as a dashed line. In some implementations, execution of a particular step (e.g., "Connect to Data") causes the corresponding block (e.g., the first block **612-1**) and/or text within the block to be visually emphasized (e.g., not grayed out or text displayed with a dark color) in the graphical user interface **600**, as illustrated in FIG. 6F. In some implementations, execution of a particular step (e.g., "Connect to Data") causes the connecting lines to be displayed as solid lines, as illustrated in FIG. 6F. In some implementations, even though the blocks **612** (e.g., steps) are displayed sequentially in the workflow **610**, a user can elect to execute certain steps while bypassing other steps.

FIG. 6C illustrates user specification of additional team members to the modeling project. In some implementations, as illustrated in FIG. 6C, the graphical user interface **600** includes an "Add" affordance **616**, which, when selected by a user, enables the user to invite other users **620** to collaborate on the modeling project. The team (e.g., including the user/Owner **618** and the other users **620**) can explore insights and deploy and manage predictive models entirely within a sever system. As an example, business scientists can collaborate directly with front office teams to make sure they are looking at all the relevant data and are factoring in domain knowledge. In some implementations, the user can also assign respective access rights (e.g., editing and/or commenting rights) to each of the other users **620**.

In some implementations, the user can initiate a new modeling project from an existing published data source. FIG. 6D illustrates the graphical user interface **600** displaying a list **622** of published data sources (e.g., data sources

240). The user can select a published data source to initiate a new modeling project. In some implementations, the user can initiate a new modeling project from a Prep flow, such as the flow illustrated in FIGS. 5A and 5B. In some implementations, the modeling project can be created from scratch, or created by combining multiple data sources.

FIG. 6E illustrates user selection of a “Spotify” data source **624** from the list **622**.

FIG. 6F illustrates, in response to user selection of a data source, the graphical user interface **600** displays a checkmark **626** (e.g., a green checkmark) inside the first block **612-1** (e.g., corresponding to the step “Connect to Data”) to indicate completion of the step. In some implementations, the graphical user interface **600** displays a name **628** of the data source (e.g., “Spotify”) as text inside the first block **612-1**.

In some implementations, in response to user selection of a data source, the graphical user interface **600** displays various options to display data fields and/or data values of the selected data source. For example, FIG. 6F illustrates a first option **629** for displaying the fields as a list, a second option **630** for displaying the fields as a table, and a third option **631** for displaying the fields as visualizations (e.g., histograms). In some implementations, user selection of the first option **629** causes the data fields to be displayed as a table **632** that includes a plurality of data rows **633**, with each of the data rows **633** corresponding to a respective data field of the selected data source, as illustrated in FIG. 6F. In some implementations, user selection of the second option **630** causes the graphical user interface **600** to display a data table **640** that includes a plurality of data columns **642**, wherein each of the data columns **642** corresponds to a respective data field of the selected data source, as illustrated in FIG. 6H. In some implementations, user selection of the third option **631** causes the graphical user interface **600** to display a respective data visualization **654** (e.g., a data value histogram) for each data field of the selected data source, as illustrated in FIG. 6L. In the example of FIG. 6L, each of the data fields has a distinct data value histogram. The vertical axis of each bar chart represents the data values corresponding to the data field, while the length of a data bar of the bar chart represents a respective count of the values.

In some implementations, a user can perform operations such as rename data fields, remove data columns, add filters, and/or add perform calculations. For example, the graphical user interface **600** can display a data prep context menu (e.g., corresponding to Prep UI **422**, which is presented as part of the user interface **600**. FIG. 4) to further clean and shape data or create a new data source entirely for input into the model, in accordance with some implementations. FIGS. **6G** and **6I** illustrate this functionality.

FIG. 6G illustrates a user interaction with (e.g., user clicking of) an arrow indicator **636** adjacent to a field name **634** corresponding to the data field “Id.” In response to the user interaction, the graphical user interface **600** displays a dropdown menu **638** that includes a “Filter” option and a “Remove” option. The “filter” option corresponds to an option to filter data values of a data field to a subset of values that are then used by a predictive model as training data for generating predictions. The “Remove” option corresponds to an option to exclude a data field (and its values) from the dataset for the modeling project (e.g., the data field and its values will not be used as training data for the predictive model).

FIG. 6I illustrates a similar user interaction, this time with a data column **642-2**, corresponding to the data field “Year” in the data table **640**. Similar to FIG. 6G, in response to the

user selection, the graphical user interface **600** displays a dropdown menu **644** that includes a “Filter” option and a “remove” option, whose functions have been discussed above with respect to FIG. 6G. FIG. 6J illustrates that, in response to user selection of the “Filter” option in FIG. 6I, a widget **646** is generated and displayed (e.g., using a widget generation module). The widget **646** identifies the data field **648** (e.g., “Year”) to which the filter is applied.

In some implementations, as illustrated in FIG. 6J, the widget **646** includes one or more filtering options **650**, such as a first option **650-1** to filter data values of the data field **648** (e.g., “Year”) according to their relative dates, a second option **650-2** to filter data values of the data field according to a range of dates, a third option **650-3** to filter data values of the data field according to a starting date, a fourth option **650-4** to filter data values of the data field according to an ending date, and a customized (e.g., special) fifth option **650-5**. In this example, the user selects the second option **650-2** to filter data values of the data field “Year” by a range of dates (the user selection is indicated by the line under the second option **650-2**).

The transition from FIG. 6J to FIG. 6K illustrates another user interaction with a slider bar icon **652**. In this example, the user slides (e.g., by holding and dragging using a mouse) the icon **652** from position A in FIG. 6J to position B in FIG. 6K. In response to the user interaction, the range of dates is modified from “Jan. 1, 1921 to Jan. 1, 2021” in FIG. 6J to “Jan. 1, 2020 to Jan. 1, 2021” in FIG. 6K.

In some implementations, once the data corresponding to the data source is in a desired state, a user can proceed to define a modeling objective. This corresponds to the step “Define Objective” (e.g., the second block **612-2**) of the workflow **610**.

To define the objective for the model, the user selects an outcome field and a modeling goal. The user can either specify a model algorithm or use a default (e.g., predetermined) model algorithm. FIGS. 6M and 6N illustrate user selection of an outcome field (e.g., a target data field) from the selected data source. In the example of FIG. 6N, each of the options that is displayed in the dropdown menu **656** corresponds to a data field in the “Spotify” data source. In FIG. 6N, the user selects data field **658** (e.g., “Popularity”) as the target field and a modeling goal (e.g., modeling objective) to “maximize” (**660**) the target field.

In some implementations, in response to user selection of the target data field, the computing device **200** (or the server **300**) determines (e.g., computes or calculates) correlations between the target data field and other data fields of the data source. FIG. 6O illustrates the determined correlations displayed as a ranked chart **662**. In this example, each data bar in the chart **662** corresponds to a field name of a data field in the “Spotify” data source and the length of a bar represents the correlation (e.g., as a percentage value) between the target data field (e.g., “Popularity”) and the respective data field. The user can select or remove/unselect fields using their respective checkboxes **664**. For example, in some circumstances, a user may unselect fields with low correlation. In some implementations, fields with 100% correlation will be automatically unselected to prevent data leakage. In some implementations, the graphical user interface **600** displays a tooltip or information icon to inform the user of why fields are removed/unselected.

In some implementations, once a modeling goal is selected (and confirmed by the user), the automatic modeling (e.g., model training, model generation, model fitting) process begins. This corresponds to the step “Train Model” (the third block **612-3**) of the workflow **610**.

In some implementations, the data source resides at a data visualization server (e.g., the third server **430**, which is the Tableau server) whereas the modeling process (e.g., model training) is performed at another server (e.g., a server that includes a predictive analytics generation module **336** or an Einstein Discovery module **414**, such as server **300**, the second server **420**, or a Tableau CRM server). In this instance, data will be seamlessly transmitted from the data visualization server to the other server for training a predictive model.

In particular, because most of the datasets (e.g., data sources) in the data visualization server are cleaned and ready to use for dashboards to inform their business, users are able to easily bring that cleaned up data into their model building experience.

In some implementations, data that is transmitted from the data visualization server (e.g., the third server **430**) to the predictive analytics server (e.g., the second server **420**) can include the target data field and its data values. In some implementations, the transmitted data can include at least a subset of data fields of the data source and their corresponding data values. For example, the subset of data fields and/or data values can be data fields and/or values that have been filtered or have a threshold correlation with the target data field. In some implementations, the transmitted data can include attributes or interpretations (e.g., user intent inferred from natural language processing) of data fields and/or values that were obtained from prior data visualization analyses, calculations. In some implementations, the transmitted data includes data that has been previously prepared for analyses or specifically prepared for model training. In some implementations, the target field and the modeling objective are selected by the user while the modeling objective is selected by the computing device **200** (or the second server **420**).

In some implementations, during the model training process, the graphical user interface **600** displays a progress bar **666** (e.g., process bar) with stops, for visualizing a real-time status of the model training. This is illustrated in FIG. **6P**. The progress bar **666** can include animations and relevant information such as time elapsed, percentage completed, identification of stages (e.g., 6-10 stages of training), and possible errors encountered.

In some implementations, the model training process can take several minutes (e.g., 5-15 minutes). During this time, the graphical user interface **600** can display a notification **667** (e.g., below the progress bar **666**) that informs a user that the user can close the browser without losing work. The user can perform manual exploration of the training data source using embedded web authoring, while waiting for the Discovery engine to complete the generation of automated insights and predictive models.

In some implementations, the graphical user interface **600** identifies one or more users **668** (e.g., the model creator and/or collaborators) that will be notified when the model is complete. In some implementations, the graphical user interface **600** displays one or more other affordances, including an affordance to add more recipients **669**, an affordance to notify one or more users via Slack, and/or a “Cancel Run Model” affordance. In some implementations, the graphical user interface **600** also displays progress indication (e.g., a time remaining), and an indication of where to find the pending model (e.g., in the user’s personal directory in the computing device **200** or in a directory located on the Cloud).

FIGS. **6Q** to **6W** illustrate evaluation of the trained model in accordance with some implementations. In some imple-

mentations, the evaluation step corresponds to the step “Model Results” (the fourth block **612-4**) of the workflow **610**. In some implementations, all the information about the model that is shown in the graphical user interface **600** is derived from data that is transmitted to the Prometheus Microservice **402** (e.g., the authoring UI **404**, or the graphical user interface **600**) from the Einstein Discovery **414** cloud.

FIG. **6Q** illustrates the graphical user interface **600** displaying a title **670** (e.g., a header) of the trained model. The title **670** can include a name (e.g., “Spotify Model”), the target field, the modeling goal (“Maximize Popularity”), and a date/time that the model was last updated. In some implementations, the graphical user interface **600** displays model metrics **672** that enable a user to evaluate the quality of a trained model. The model metrics **672** can include a first metric **672-1** “Overview,” which provides a summary of the model’s validation results, a second metric **672-2** “Model Evaluation,” which provides information about a model’s performance, gains and lift, cross-validation results, and coefficient values, a third metric **672-3** “Threshold Evaluation,” which helps a user optimize a threshold value for a model, and a fourth metric **672-4** “Prediction Evaluation,” which enables a user to interact with the model metrics and see how they stack up against actual outcomes.

In some implementations, and as illustrated in FIG. **6Q**, the graphical user interface **600** displays details **674** of the trained model. This can include a date and/or time at which the model was created, a date and/or time that the model was updated, an identification of the user who created the model (e.g., who initiated the modeling project), an identification of the modeling algorithm (e.g., generalized linear model or “GLM”), and/or a type of statistical analysis (e.g., “Regression”) that was applied for the model training.

FIG. **6Q** also illustrates, in some implementations, the title **670**, the metrics **672**, and the model details **674** are displayed in a side pane **676** of the graphical user interface **600**. In some implementations, the graphical user interface **600** also displays an affordance (e.g., on the side pane **676** or elsewhere) that, when selected, causes the trained model to be deployed.

In some implementations, each of the model metrics **672** is a user-selectable affordance. For example, FIG. **6Q** illustrates in response to user selection of the first metric **672-1** “Overview,” the graphical user interface **600** displays an at-a-glance summary **675** of the model’s validation results. FIGS. **6R**, **6S**, and **6T** illustrate in response to user selection of the second metric **672-2** “Model Evaluation,” the graphical user interface **600** displays navigation tabs **678** (e.g., a first navigation tab **678-1** “Overall Performance,” a second tab **678-2** “Cross Validation,” and a third tab **678-3** “Coefficients”), which, when selected by the user, cause respective information (e.g., parameters) about the model evaluation metric to be displayed.

FIGS. **6U** and **6V** illustrate selection of the third metric **672-3** and the fourth metric **672-4**.

FIGS. **6W** to **6AA** illustrate model deployment in accordance with some implementations. This corresponds to the “Deploy Model” step **612-5** of the workflow **610**. In some implementations, Once the Model is deployed, the “Deploy Model” block **612-5** becomes available (e.g., selectable).

Deploying a model means using the model that has been trained to make new predictions. In some implementations, deploying the model comprises taking the trained model (e.g., which exists in the second server **420**, such as in the Einstein Discovery module **414**) and enabling it to start generating predictions.

Selecting “Deploy Model” activates the already trained model object and instantiates the API endpoints that allow it to be called for training. Stated another way, deploying a model enables it to start generating predictions as an API endpoint (e.g., the third API **338-3**) in a server, such as the third server **430** (e.g., Tableau server), a Salesforce server, or other servers. Predictions can be consumed in data and visualizations, tracked for performance and data drift, and integrated into the data catalog all inside the data visualization server (e.g., the third server **430**, Tableau server).

In some implementations, once the model is trained, the trained model is stored as an immutable object in the cloud (e.g., on a server system such as the ML platform) (E.g., the server **300** or the second server **420**). The trained model can be deleted. In some implementations, if the user re-runs the training flow, a new object is created, but since this is remote from where the user is running the training user interface (e.g., graphical user interface **600**), the model itself is not stored locally in memory.

In some implementations, for the “Deploy Model” step of the workflow, the graphical user interface **600** displays an “Overview” tab **680** and a “Next Steps” tab **682**.

In some implementations, in response to user selection of the “Overview” tab, the graphical user interface **600** displays information such as a deployment status of the location, a location of the model, a deployment history, as illustrated in FIGS. **6W**, **6X**, and **6Y**. In some implementations, the graphical user interface **600** displays a lineage tree to track where the model is being used. For example, the lineage tree can identify Sheets (e.g., worksheets), Dashboards, Calcs, and other files where the model is used.

In some implementations, in response to user selection of the “Next Steps” tab **682**, the graphical user interface **600** includes a Quick Start guide with instructions on how the new model can be used. These may include Web Authoring, Links to Prep Flow, Auto connect to workbooks in Einstein Discovery, Autogenerating Tableau calcs, and other helpful guides. This is illustrated in FIGS. **6Z** and **6AA**.

In some implementations, the models are created in, physically stored in, and generate predictions from the second server **420** (e.g., the Tableau CRM server), but will appear in the Tableau server (e.g., the third server **430**) as a new object type and is fully supported in the Tableau Catalog. The model itself lives in the Tableau server as a part of its modeling project, which contains all model training steps as well as tracking for observation of the model in production. In some implementations, data from the modeling projects are saved as standalone data, or are tied to existing published data sources. Because the models in the Tableau catalog are connected to the data sources that were used to train them, all modeling decisions and data lineage can be tracked and audited (e.g., traceable). In some implementations, the catalog also tracks where models are being consumed in Prep flows and workbooks. Any user with access to the modeling project can return after the model is in production to audit the project and review data and model choices and assumptions.

In some implementations, during the execution of the workflow **610**, the computing device (e.g., and/or the server system **300**, **410**, **420**, or **430**) determines artifacts corresponding to each of the steps of the workflow **610**. As used herein, an “artifact” refers an item generated and exchanged by human or machine actions across an end-to-end automated data science (ML) workflow that comprises preparation, analysis, deployment, and communication stages (e.g., the steps as discussed with respect to FIG. **6B**). For example, the model, the descriptive statistics about the model perfor-

mance, correlations between the target data field and other data fields of the data source, data changes that are made (e.g., removal of one or more data columns from the data source, filtering data rows, etc.) are artifacts generated in the predictive modeling project. In some implementations, the computing device **200** (e.g., and/or the server system **300**, **410**, **420**, or **430**) stores all the artifacts of a predictive modeling project in a single location (e.g., on the computing device or the server system), enabling a user can easily review (e.g., audit) model decisions and understand how data drove the model as a whole and contributed to individual predictions.

In some implementations, the computing device **200** (or the second server system **420**) generates one or more views for visualizing artifact interactions across the ML workflow (e.g., a data analysis workflow pipeline). Details of visualizing artifact interactions are disclosed in U.S. Provisional Patent Application No. 63/242,012, filed Sep. 8, 2021, entitled “Visual Tracing and Editing of Machine Learning Models,” which is incorporated by reference herein in its entirety.

Thus, in accordance with some implementations, the disclosed systems and user interfaces align with the design principles of integrated, iterative, traceable, guided, collaborative, and fluid (e.g., liquid) ML workflows.

Flowcharts

FIGS. **7A-7D** provide a flowchart of a method **700** for generating predictive analytics based on no-code machine learning (ML) models according to some implementations. The method **700** is also called a process.

The method **700** is performed (**702**) at a computing device **200** that has a display **208**, one or more processors (e.g., CPU(s)) **202**, and memory **214**. The memory **214** stores (**704**) one or more programs configured for execution by the one or more processors **202**. In some implementations, the operations shown in FIGS. **5A** to **5B** and FIGS. **6A** to **6AA** correspond to instructions stored in the memory **214** or other non-transitory computer-readable storage medium. The computer-readable storage medium may include a magnetic or optical disk storage device, solid state storage devices such as Flash memory, or other non-volatile memory device or devices. The instructions stored on the computer-readable storage medium may include one or more of: source code, assembly language code, object code, or other instruction format that is interpreted by one or more processors. Some operations in the method **700** may be combined and/or the order of some operations may be changed.

The computing device **200** displays (**706**), in a user interface (e.g., graphical user interface **600**), a workflow (e.g., a workflow interface or a workflow pipeline) (e.g., the workflow **610** as illustrated in FIGS. **6B**, **6D**, and **6F**), which includes a plurality of steps (e.g., the steps of the workflow are represented as blocks **612**). In some implementations, the plurality of steps can include steps such as “connect to data” (e.g., select a data source), “objective definition” (e.g., define a modeling objective for a target data field of a selected data source), “model training (e.g., model generation, model optimization), “display model results,” and “deploy model.” In some implementations, alternative names of steps are used to specify the same or equivalent functions.

In some implementations, each of the steps of the workflow is (**708**) a user-selectable element in the user interface.

Responsive to user selection of a first step (e.g., “Connect to Data”) of the plurality of steps, the computing device displays (**710**) a list of data sources (e.g., the data sources **240**).

The computing device **200** receives (**712**) user selection of a first data source (or dataset) of the data sources (e.g., the “Spotify” data source, as illustrated in FIG. **6E**).

In some implementations, after receiving user selection of the data source, the computing device **200** receives further user input to clean up/prepare the data. For example, the user can specify filters to filter data fields of the data source to a subset of data fields, or filter data values of one or more data fields of the data source to a subset of data values, which will be used for model training.

The computing device **200** receives (**714**) user input specifying a target data field (e.g., a target variable, an outcome variable, or an outcome field) from the first data source (e.g., the data field **658** “Popularity” as illustrated in FIG. **60**, or any data field in the data source, such as “Tempo,” “Release Date,” “Sales,” or “Profit”) and a modeling objective (e.g., modeling goal) for the target data field (e.g., maximize, minimize, increase, decrease, or classify the target field).

In response to (**716**) the user input, the computing device **200** automatically executes (**718**) (e.g., generates or trains) (e.g., without user intervention) a model (e.g., a predictive model) to predict a first outcome for the target data field based on the modeling objective. For example, in some implementations, the model optimizes the modeling objective for the target data field. In some implementations, automatically executing the model includes computing statistics for evaluating performance of the model.

The computing device displays (**722**) results of the model.

In some implementations, the results of the model are displayed (**726**) on a side pane of the user interface (e.g., a side pane **676** of the interface **600**, as illustrated in FIG. **6Q**).

In some implementations, displaying the results of the model includes displaying (**728**) one or more of: a model title, a date of generation of the model, a date of update of the model, and a version of the model.

In some implementations, the computing device **200** also displays an affordance (e.g., a user-selectable element) to deploy the model.

In some implementations, displaying the results of the model includes (**730**) displaying a plurality of metrics of the model (e.g., metrics **672** such as “Overview,” “Model Evaluation,” “Threshold Evaluation,” and/or “Prediction Evaluation,” as illustrated in FIG. **6Q**).

The computing device **200** receives (**732**) user input to deploy the model.

In accordance with the user input, the computing device **200** deploys (**734**) the model.

For example, deploying the model can include using the trained model to make new predictions. In some implementations, deploying the model comprises taking the trained model (e.g., that exists in the server **420** (e.g., in the Einstein Discovery module **414**) and enabling it to start generating predictions as an API endpoint (e.g., the third API **338-3**) in a server such as the third server **430** (e.g., Tableau server), a Salesforce server, or other servers.

In some implementations, once a model is trained, the trained model is stored as an immutable object in the cloud (e.g., on a server system). The trained model can be deleted, and if the user re-runs the training flow, a new object is created. Because the trained model is located remotely from where the user is running the training user interface, the model itself is not stored locally in memory.

In some implementations, all information about the model that is shown in the graphical user interface **600** is derived from data that is transmitted back to the Prometheus Microservice **402** (e.g., the authoring UI **402** or the graphical user

interface **600**) from the second server **420**, which houses the Einstein Discovery module **414**. Selecting “Deploy Model” activates the already trained model object and instantiates the API endpoints that allow it to be called for training.

In some implementations, deploying the model comprises sending (**736**) the model to a plurality of users via a plurality of modes of communication (e.g., by instant messaging, by Slack channels, by email, via a messaging application, or via a data visualization application).

In some implementations, the first step (e.g., step **612-1**) corresponds (**738**) to a first user-selectable element. In response to user selection of the first element, the computing device **200** displays (**740**) the first element in a visually distinct manner from other elements corresponding to other steps in the workflow. For example, in some implementations, at the start of the modeling project, each of the steps (e.g., the blocks **612**) is grayed out (inactive) in the graphical user interface **600**. Execution of a particular step causes that step to be visually emphasized (e.g., activated) in the graphical user interface **600**. In some implementations, while the workflow steps are displayed in a linear manner, as illustrated in FIG. **6B**, a user can elect to execute certain steps while bypassing other steps of the workflow **610**.

In some implementations, after receiving the user input specifying the target data field and the modeling objective, the computing device **200** updates (**742**) the first element to indicate completion of the first step (e.g., by adding a checkmark (e.g., a green checkmark) in the block **612** corresponding to the first element).

In some implementations, in response to (**744**) the user input specifying the target data field, the computing device **200** determines (**746**) one or more second data fields from the data source and a respective correlation between the target data field and each of the one or more second data fields. The computing device **200** displays (**748**) the respective correlations as a ranked bar chart on the user interface. This is illustrated in FIG. **60**.

In some implementations, the computing device **200** receives (**750**) user selection of a first metric **672** of the plurality of metrics. In response to the user selection, the computing device displays (**752**) a plurality of navigation tabs **678** on the user interface. User selection of a respective navigation tab causes respective information (e.g., parameters) about the first metric to be displayed.

In some implementations, the computing device **200** displays (**720**) a progress bar (e.g., progress bar **666**) for visualizing progress (e.g., real-time progress) of the model generation. In some implementations, the progress bar includes information such as time elapsed, percentage completed, identification of stages of the model generation, and possible errors encountered.

In some implementations, the computing device **200** receives (**754**) user specification of a plurality of users to which the model is to be deployed. In some implementations, the user specification includes a respective level of access (e.g., rights to edit the model, view the model only, and/or re-share the model with other users).

In some implementations, the user selection of the first step initiates (**756**) a predictive modeling project. The computing device **200** stores (**758**) all artifacts of the predictive modeling project on the computing device. In some implementations, the artifacts are stored on a server system (e.g., server **300**, **410**, **420**, and/or **430**), which is communicatively connected with the computing device.

In some implementations, the computing device **200** receives (**760**) user modification of a first artifact of the artifacts. In accordance with the user modification, the

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computing device 200 automatically executes (762) (e.g., iterates) the model to predict an updated outcome for the target data field based on the modeling objective and the first artifact. In some implementations, the computing device 200 (or the server 300, 420, or 430) associates every iteration of the model with a corresponding version number for the model and/or a date/time stamp. The computing device 200 (or the server 300, 420, or 430) further stores all artifacts corresponding to that version number, thereby providing an auditable trail for all data and modeling choices.

Each of the above identified executable modules, applications, or sets of procedures may be stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing a function described above. The above identified modules or programs (i.e., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various implementations. In some implementations, the memory 214 stores a subset of the modules and data structures identified above. Furthermore, the memory 214 may store additional modules or data structures not described above.

The terminology used in the description of the invention herein is for the purpose of describing particular implementations only and is not intended to be limiting of the invention. As used in the description of the invention and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

The foregoing description, for purpose of explanation, has been described with reference to specific implementations. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The implementations were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to best utilize the invention and various implementations with various modifications as are suited to the particular use contemplated.

What is claimed is:

1. A method of generating predictive analytics based on no-code machine learning (ML) models, comprising:
 - at a computing device having a display, one or more processors, and memory storing one or more programs configured for execution by the one or more processors, including instructions to perform:
 - displaying, in a user interface, a workflow that includes a plurality of steps of the workflow; and
 - responsive to receiving user selection, via the user interface, of a first step of the plurality of steps, initiating a predictive modeling project, including:
 - displaying, in the user interface, a list of data sources;
 - receiving, via the user interface, user selection of a first data source of the data sources;

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receiving, via the user interface, first user input specifying a target data field from the first data source and a modeling objective for the target data field;

in response to receiving the first user input specifying the target data field and the modeling objective for the target data field:

determining, by the computing device without user input, (i) one or more data fields from the data source and (ii) a respective correlation between the target data field and each of the one or more data fields, and displaying the one or more data fields in the user interface;

receiving, via the user interface, second user input (a) selecting at least one of the one or more data fields to include in the predictive modeling project or (b) deselecting at least one of the one or more data fields to exclude from the predictive modeling project;

in accordance with receiving the second user input: automatically executing a model to predict a first outcome for the target data field based on the modeling objective;

while automatically executing the model, obtaining a plurality of artifacts for the predictive modeling project, wherein a respective artifact of the plurality of artifacts is an item generated and exchanged by human or machine actions across an end-to-end automated ML workflow that comprises preparation, analysis, deployment, and communication stages;

generating and displaying results of the model in the user interface, including generating and displaying an artifact dependency view that shows dependency relationships between the plurality of artifacts, including, in response to user interaction in the user interface with a particular artifact displayed in the artifact dependency view, highlighting in the user interface the associated dependencies of the particular artifact, wherein:

the artifact dependency view includes (i) a plurality of visual representations corresponding to the plurality of artifacts and (ii) a plurality of connectors, wherein a respective connector of the plurality of connectors connects two visual representations of the plurality of visual representations, corresponding to two artifacts of the plurality of artifacts that have a dependency relationship with each other; and the plurality of visual representations includes a first visual representation corresponding to a human-generated artifact, a second visual representation corresponding to a machine-generated artifact, and a third visual representation corresponding to an artifact with a missing source, wherein displaying the artifact dependency view includes displaying the first visual representation, the second visual representation, and the third visual representation with different visual characteristics;

updating a metadata database of a server to include the plurality of artifacts;

receiving, via the user interface, third user input to deploy the model to a plurality of subjects; and in accordance with receiving the third user input, deploying the model to a plurality of devices

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associated with the plurality of subjects via a communication interface of the computing device, the plurality of devices including a first device executing a messaging application and a second device executing a data visualization application.

2. The method of claim 1, wherein each of the plurality of steps of the workflow is a user-selectable element in the user interface.

3. The method of claim 2, wherein the first step of the workflow corresponds to a first user-selectable element, the method further comprising:

in response to user selection of the first user-selectable element corresponding to the first step of the workflow, displaying the first user-selectable element in a visually distinct manner from other user-selectable elements corresponding to other steps of the plurality of steps of the workflow.

4. The method of claim 3, further comprising:

after receiving the first user input specifying the target data field and the modeling objective, updating the first user-selectable element to indicate completion of the first step of the workflow.

5. The method of claim 1, wherein the results of the model are displayed on a side pane of the user interface.

6. The method of claim 1, wherein displaying the results of the model includes displaying one or more of: a model title, a date of generation of the model, a date of update of the model, and a version of the model.

7. The method of claim 1, wherein:

displaying the results of the model includes displaying a plurality of metrics of the model; and

the method further comprises:

receiving user selection of a first metric of the plurality of metrics; and

in response to the user selection of the first metric of the plurality of metrics, displaying a plurality of navigation tabs in the user interface, wherein user selection of a respective navigation tab causes respective information about the first metric to be displayed.

8. The method of claim 1, further comprising:

displaying a progress bar for visualizing progress of the model execution.

9. The method of claim 1, further comprising receiving user specification of a plurality of users to which the model is to be deployed.

10. The method of claim 1, further comprising:

storing all artifacts generated from the predictive modeling project on the computing device.

11. The method of claim 10, further comprising:

receiving user modification of a first artifact of the artifacts; and

in accordance with the user modification, automatically executing the model to predict an updated outcome for the target data field based on the modeling objective and the first artifact.

12. The method of claim 1, wherein deploying the model comprises sending the model to a plurality of users via a plurality of modes of communication.

13. The method of claim 1, further comprising:

after determining the respective correlation between the target data field and each of the one or more data fields:

displaying the respective correlation as a ranked bar chart in the user interface, wherein the ranked bar chart includes a plurality of each of the data bars, each of the data bars corresponding to a field name of a respective data field of the one or more data fields; and

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receiving the second user input includes receiving user selection or deselection, from the ranked bar chart, of at least one data bar corresponding to the respective data field of the one or more data fields.

14. The method of claim 1, further comprising storing the plurality of artifacts on the server.

15. The method of claim 1, wherein generating and displaying the results of the model in the user interface includes:

generating and displaying an artifact origin view that identifies a first subset of the plurality of artifacts that are human-generated and a second subset of the plurality of artifacts, distinct from the first subset of the plurality of artifacts, that are machine-generated; and the artifact origin view comprises an alluvial diagram having a first end, a second end, and a plurality of first connectors, the first end including a plurality of second visual representations that identifies a human-generated source or a machine-generated source, the second end includes a plurality of third visual representations that each corresponds to a respective artifact of the plurality of artifacts, and a respective connector connects a respective second visual representation to a respective third visual representation.

16. The method of claim 1, wherein generating and displaying the artifact dependency view includes:

in response to receiving a user interaction with a respective visual representation, corresponding to a respective artifact of the plurality of artifacts, in the artifact dependency view:

displaying an information window that includes information about the respective artifact.

17. A computing device for generating predictive analytics based on no-code machine learning (ML) models, comprising:

a display;

one or more processors; and

memory coupled to the one or more processors, the memory storing one or more programs configured to be executed by the one or more processors, the one or more programs including instructions for:

displaying, in a user interface, a workflow that includes a plurality of steps; and

responsive to receiving user selection, via the user interface, of a first step of the plurality of steps, initiating a predictive modeling project, including:

displaying, in the user interface, a list of data sources;

receiving, via the user interface, user selection of a first data source of the data sources;

receiving, via the user interface, first user input specifying a target data field from the first data source and a modeling objective for the target data field;

in response to receiving the first user input specifying the target data field and the modeling objective for the target data field:

determining, by the computing device without user input, (i) one or more data fields from the data source and (ii) a respective correlation between the target data field and each of the one or more data fields, and displaying the one or more data fields in the user interface;

receiving, via the user interface, second user input (a) selecting at least one of the one or more data fields to include in the predictive modeling project

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or (b) deselecting at least one of the one or more data fields to exclude from the predictive modeling project;

in accordance with receiving the second user input: automatically executing a model to predict a first outcome for the target data field based on the modeling objective;

while automatically executing the model, obtaining a plurality of artifacts for the predictive modeling project, wherein a respective artifact of the plurality of artifacts is an item generated and exchanged by human or machine actions across an end-to-end automated ML workflow that comprises preparation, analysis, deployment, and communication stages;

generating and displaying results of the model in the user interface, including generating and displaying an artifact dependency view that shows dependency relationships between the plurality of artifacts, including, in response to user interaction in the user interface with a particular artifact displayed in the artifact dependency view, highlighting in the user interface the associated dependencies of the particular artifact, wherein:

the artifact dependency view includes (i) a plurality of visual representations corresponding to the plurality of artifacts and (ii) a plurality of connectors, wherein a respective connector of the plurality of connectors connects two visual representations of the plurality of visual representations, corresponding to two artifacts of the plurality of artifacts that have a dependency relationship with each other; and

the plurality of visual representations includes a first visual representation corresponding to a human-generated artifact, a second visual representation corresponding to a machine-generated artifact, and a third visual representation corresponding to an artifact with a missing source, wherein displaying the artifact dependency view includes displaying the first visual representation, the second visual representation, and the third visual representation with different visual characteristics;

updating a metadata database of a server to include the plurality of artifacts;

receiving, via the user interface, third user input to deploy the model to a plurality of subjects; and

in accordance with the third user input, deploying the model to a plurality of devices associated with the plurality of subjects via a communication interface of the computing device, the plurality of devices including a first device executing a messaging application and a second device executing a data visualization application.

18. The computing device of claim 17, wherein the first step included in the workflow corresponds to a first user-selectable element in the user interface, and the one or more programs further include instructions for:

in response to user selection of the first user-selectable element corresponding to the first step included in the workflow, displaying the first user-selectable element in a visually distinct manner from other user-selectable elements corresponding to other steps of the plurality of steps of the workflow.

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19. The computing device of claim 18, wherein the one or more programs further include instructions for:

after receiving the first user input specifying the target data field and the modeling objective, updating the first user-selectable element to indicate completion of the first step included in the workflow.

20. A non-transitory computer-readable storage medium storing one or more programs, the one or more programs comprising instructions, which when executed by a computing device, cause the computing device to perform operations comprising:

displaying, in a user interface, a workflow that includes a plurality of steps; and

responsive to receiving user selection, via the user interface, of a first step of the plurality of steps, initiating a predictive modeling project, including:

displaying, in the user interface a list of data sources; receiving, via the user interface, user selection of a first data source of the data sources;

receiving, via the user interface, first user input specifying a target data field from the first data source and a modeling objective for the target data field;

in response to receiving the first user input specifying the target data field and the modeling objective for the target data field:

determining, by the computing device without user input, (i) one or more data fields from the data source and (ii) a respective correlation between the target data field and each of the one or more data fields, and displaying the one or more data fields in the user interface;

receiving, via the user interface, second user input (a) selecting at least one of the one or more data fields to include in the predictive modeling project or (b) deselecting at least one of the one or more data fields to exclude from the predictive modeling project;

in accordance with receiving the second user input: automatically executing a model to predict a first outcome for the target data field based on the modeling objective;

while automatically executing the model, obtaining a plurality of artifacts for the predictive modeling project, wherein a respective artifact of the plurality of artifacts is an item generated and exchanged by human or machine actions across an end-to-end automated ML workflow that comprises preparation, analysis, deployment, and communication stages;

generating and displaying results of the model in the user interface, including generating and displaying an artifact dependency view that shows dependency relationships between the plurality of artifacts, including, in response to user interaction in the user interface with a particular artifact displayed in the artifact dependency view, highlighting in the user interface the associated dependencies of the particular artifact, wherein:

the artifact dependency view includes (i) a plurality of visual representations corresponding to the plurality of artifacts and (ii) a plurality of connectors, wherein a respective connector of the plurality of connectors connects two visual representations of the plurality of visual representations, corresponding to two artifacts of the plurality of artifacts that have a dependency relationship with each other; and

the plurality of visual representations includes a first visual representation corresponding to a human-generated artifact, a second visual representation corresponding to a machine-generated artifact, and a third visual representation 5 corresponding to an artifact with a missing source, wherein displaying the artifact dependency view includes displaying the first visual representation, the second visual representation, and the third visual representation with 10 different visual characteristics;

updating a metadata database of a server to include the plurality of artifacts;

receiving, via the user interface, third user input to deploy the model to a plurality of subjects; and 15 in accordance with the third user input, deploying the model to a plurality of devices associated with the plurality of subjects via a communication interface of the computing device, the plurality of devices including a first device executing a messaging appli- 20 cation and a second device executing a data visualization application.

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